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Review

A review of the present and future utilisation of FRP composites in the civil infrastructure with reference to their important in-service properties

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ABSTRACT

The paper discusses the development of the advanced polymer composite material applications in the building and civil/structural infrastructure over the past three to four decades. It endeavours to identify and prioritise the important in-service research areas which are necessary to improve the understanding of the behaviour of FRP materials and FRP structural components. The paper demonstrates the types of structures which have been developed from the FRP composite material and the most advantageous way to employ composites in civil engineering. The material has extraordinary mechanical and important in-service properties which when combined with other materials are utilised to improve the stiffness/strength, durability, the whole-life cost benefit and the environmental impact. The paper concludes by summarising key successes of the advanced polymer composite in the civil infrastructure and suggests areas in which, if they are employed innovatively, FRP composites could be used with great advantage.

1. Introduction

For more than 30 years following the Second World War the construction industry showed a lack of investment in research and development and consequently potential material investors, in the technological revolution in materials and in their processing techniques, were being explored in other sectors of the manufacturing industry and inevitably the construction industry was bypassed, Latham [134] and Egan [61]. Nevertheless, notwithstanding the criticisms by these authors there is evidence in the late 1970s and into the 1980s of an interest by the research departments of universities, research institutes and a few civil engineering consultants in advanced polymer composite (APC) materials; these materials consist of high-strength and stiffness fibres protected by a high-performance thermosetting polymer. The early research and development and the innovations in structural and civil engineering APC systems was instrumental in the current interest, throughout the world, in the use of APC in the civil engineering industry.

The polymer composite derives its mechanical characteristics wholly from those of the fibre and the quality of the fibre/matrix interface, therefore the most important properties required of the polymer is its physical and in-service characteristics. High-performance thermosetting resins are required to provide specific properties in highly demanding environments. These matrices must possess high dimensional stability at elevated temperatures and thermal resistance, low water absorption, good chemical resistance, high mechanical strength, excellent stiffness and high compressive strength. This combination of properties is essential for advanced composites to be utilised in the construction industry, but due to the increase in cross-linking density observed during polymerisation, conventional thermosetting matrices are considered to be brittle and this limits the damage tolerance of the composite, O'Brien [176], Hollaway [98].

Before discussing the current and future composite structural systems used in the civil infrastructure it is important to discuss the characteristics of the material which make them attractive in some areas of construction and other characteristics which require to be improved before full confidence in the material is achieved. This paper, therefore, will be divided into two parts.

Part A will examine the in-service and physical properties of polymers and composites for utilisation in civil engineering. These characteristics are fundamental for a successful structural system to be used in the civil infrastructure.

Part B will demonstrate how these unique characteristics of APCs can be used to form,

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- ‘All-FRP-composite’ structures.
- Combined with other engineering materials to improve the stiffness, strength and durability of the overall composite structural member.
- Future generations of FRP structural members associated with the construction industry.

The mechanical properties of the component parts of the composite are clearly important but this area has been well documented, Hollaway and Head [90], Hull and Clyne [107], Kim [125] and only a brief discussion will be included in Part A as (a) mechanical properties will be affected by the in-service properties over time and (b) for completeness. Likewise, a brief discussion will be given of the mechanical properties of the FRP composite; typical mechanical values are given in Appendix A.

In civil engineering the APC is generally referred to as the fibre-reinforced polymer (FRP) composite; this description will be used throughout this paper.

2. Part A: the important physical and in-service properties of thermosetting polymers used in the civil infrastructure

2.1. Introduction

The FRP engineering structural composites must possess not only sufficient strength and stiffness properties to resist the full superimposed and self-weight loads to which the structure is exposed but also the relevant in-service and physical characteristics required to function in the aggressive and sometimes hostile environments encountered in the construction industry; these latter characteristics are clearly just as important as the mechanical properties. The greater the degradation of structures over time the lower will be their load carrying capacity. Consequently, the most important properties of the matrix (the polymer), which protects the load carrying fibre component of the composite, are its physical and in-service characteristics.

The vinyl-esters, the epoxies and the polyesters are the thermosetting matrices which are utilised for composite structural members in the civil infrastructure; all are crossed linked. A wide range of amorphous and crystalline polymer materials (an amorphous and a crystalline polymer are those in which there is a random order of their atoms and those in which there is an orderly repeating pattern of their atoms, respectively) can be used to form fibres. In the construction industry the three fibres which are invariably used are the glass, the aramid and the carbon fibres. The basic mechanical properties of the component parts of the composites, their interaction and the techniques for the manufacture of the fibres and the composite materials have been discussed many times in a number of publications, Kim [125], Hollaway and Head [90], Karbhari [118], Hollaway [98] and will not be dealt with here. However, the physical and in-service characteristics of the component parts of the FRP composite will be discussed in the subsequent sections. These characteristics are of primary importance in relation to the durability of the polymer and hence of the FRP composite.

2.2. Polymerisation

It is essential that polymers are manufactured correctly for them to perform their in-service functions efficiently. Polymerisation is a process of bonding together repeating molecular building blocks, known as monomers, through a variety of reaction mechanisms to form large chainlike or network molecule of relatively high molecular mass known as a polymer. At least one hundred and often thousands of monomer molecules must be combined to form a product that has certain physical properties such as high-modulus of elasticity and high tensile strength values or has the ability to form fibres. There are two classes of polymerisation, these are:

- *Addition polymerisation* is a process in which monomers react to form a polymer without the formation of by-products. Addition polymerisation is usually undertaken in the presence of a catalyst, which in certain cases controls the structural properties of the polymer. In this process monomers are dissolved in a solvent that is later removed. The monomers quickly combine by an addition reaction without losing any atoms, so that the polymer has the same basic formula as the monomer.
- *Condensation polymerisation* is a slower stepwise reaction. It results in the loss of atoms or groups of atom as by-products of the linking monomers. Most condensation polymerizations are of a kind of copolymerization, usually consisting of two or more types of monomers. The number of monomers in a polymer determines the degree of polymerization of the polymer. When the number of monomers is high, the compound is said to have a high degree of polymerization and is called a high polymer.

As mentioned earlier thermosetting resins are crossed linked polymers, in which their molecular structure is a network. These resins are formed under the influence of heat and once formed they do not melt or soften upon reheating, and do not dissolve in solvents; they can be made by either addition or condensation polymerisation. It is essential that the correct mix ratio is obtained between the epoxy resin and the curing agent to ensure that a complete reaction does take place as the curing agent molecules ‘co-react’ with the thermosetting resin molecules in a fixed ratio. If the mix is not in the correct proportions, un-reacted resin or curing agent will remain within the matrix, and this will affect the final properties of the polymer after cure.

There are two procedures which are used to polymerise a thermosetting polymer for the civil engineering industry, these are:

- *The cold cured systems* where the polymer is cured (polymerised) at ambient temperature on site, generally in the region of 10–30 °C; the lower the curing temperature the longer is the cure time. It is advisable with cold cure resins to provide a post cure with a higher temperature over an extended period of time. This regrettably is not generally done on civil engineering site.
- *The hot cured system* where the polymerisation is performed in a factory environment at elevated temperatures of the order of 130 °C; this is generally an automated production procedure.

The cold and hot cured resins have different formulations, consequently, a hot cured system cannot be polymerised using a cold curing agent and vice versa. Attention must be given to the site temperature when using the cold cure polymers; the environmental temperature under working conditions should be some 20° C below the *glass transition temperature* (T_g).

2.3. Temperature

The influence of temperature on polymers can be separated into two effects:

- short-term and
- long-term.

The short-term effect is generally physical and is reversible when the temperature returns to its original state, whereas the long-term effect is generally dominated by chemical change and is not reversible; this effect is referred to as aging. As the temperature varies all properties of the polymer will change, consequently, to fully characterise the temperature dependent material, properties should be measured over a range of temperatures. To study one or more of the properties as a function of temperature, a thermal analyser is used; it scans property change over a wide temperature range. The differential scanning calorimeter (DSC) is used to undertake these measurements. The heat input and temperature rise for the material under test are compared with those for a standard material both subjected to a controlled temperature programme. The general principles of DSC are outlined in ISO standard, ISO 11357-1 [1997] for polymers and it contains a bibliography giving references to more detailed information.

Particular cases of the effects of temperature on polymers are: (i) the glass transition temperature and their melting point, (ii) their thermal expansion, (iii) their thermal conductivity, and (iv) the effect of ultraviolet light although this is not strictly a temperature property.

2.3.1. Glass Transition T_g and melting point T_m

The temperature below which the physical properties of an amorphous or an amorphous/crystalline polymer vary in a manner similar to that of a solid phase (brittle or glassy state) and above which it behaves in a manner similar to that of a liquid (rubbery state) is known as the glass transition temperature (T_g). More specifically, the glass transition state is a pseudo second order phase transition in which a supercooled melt returns on cooling to a glassy structure. The T_g is usually associated with the wholly amorphous or the amorphous/crystalline polymer (the epoxies used in construction come under the latter class of material) in which it changes from the solid phase to the rubbery state (or vice versa) gradually over a finite temperature range and the T_g is the midpoint of this range. As polymers below the T_g, are rigid, they have both stiffness and strength, but above the T_g, the amorphous (or amorphous/crystalline) polymers are soft elastomers or viscous liquids, and have no stiffness or strength. The T_g of crystalline (thermoplastic) polymers are more complex than those of the amorphous/crystalline (thermosetting) polymers because, in addition to a melting temperature T_m, which takes place over a range of a few degrees and above which all their crystalline structure disappears they also have a second but lower value T_g below which they become rigid and brittle. Slightly different numerical values of the T_g may be quoted in the literature depending upon the testing technique used. There are two methods which may be used, these are the Dynamic Mechanical Thermal Analysis (DMTA), [ISO/CD standard 6721-11 (2001)] and the Differential Scanning Calorimetry (DSC) ISO standard, ISO 11357-1 [1997].

All physical properties of thermosetting polymers depend upon intermolecular cross-links for their strength and as stated above as the temperature nears its (T_g) value the polymer will begins to soften. The temperature at which this happens depends upon the detailed chemical structure of the polymer. The hot cured polyesters, vinyl-esters and epoxies all begin to weaken and break down at above 200° C, the cold cured polymers will have a lower T_g value than that of the hot cured polymers. However, the T_g of *some* low temperature (ambient cured) moulded composites, can be increased in value by further post curing the polymer at a higher temperature but for any specific cold cure thermosetting polymer there is a maximum value of the T_g, irrespective of the post cure temperature value.

2.3.2. The thermal expansion (CTE)

The CTE is the change in length per unit rise in temperature. The CTEs of polymer materials are of the order of 100×10^{-6}, consequently, they are an order higher than those of the conventional civil engineering materials. The CTEs vary with temperature ranges and are calculated as the slope of the secant line of the thermal expansion curve between the reference temperature (generally the normal environmental temperature) and the temperature of interest. The CTE of thermosetting polymers is influenced mainly by the degree of the cross-linking of the molecules of the material and the overall stiffness of the units between the cross-linkages. *This property must be considered in structural design when joining a polymer or polymer composite to a dissimilar material.*

2.3.3. The Thermal conductivity

The thermal conductivity is a measure of the ease with which temperature is transmitted through a material. The thermal conductivity of all polymers is low; consequently, they are good heat insulators. To reduce the thermal conductivity of a polymer further the material can be used in the form of a foam. If the value is to be increased, metallic fillers can be added to the resin at the time of polymerization.

2.3.4. Fire resistance

The polymer component of the composite used in the civil engineering industry is an organic material and is composed of carbon, hydrogen and nitrogen atoms; these materials are flammable to varying degrees. Consequently, a major concern for the construction engineer using polymers is the problem associated with fire. Most building structures must satisfy the requirements of building codes relating to the behaviour of structures in a fire. A measure of fire ratings for buildings refers to the time available in a fire before the structure collapses. However, the major health hazard derived from polymer and composites in a fire accident is generated from the toxic combustion products produced during burning of materials. The degree of toxicity generated depends on the phase of burning of the fire including: oxidative pre-ignition, flaming combustion or fully developed combustion and ventilation controlled fires. Smoke toxicity plays an important role during fire accidents in buildings, where the majority of people die from smoke inhalation.

The basic approaches to reduce the fire hazards of polymers are:

(a) To extinguish the fire, to control the fire, or to provide exposure protection for structures on site, by:
  - a sprinkler system,
  - a foam system,

(b) To introduce additives into resin formulations, by:
  - incorporating halogens into resins formulations (e.g. fluorine, chlorine, bromine and iodine family of chemicals),
  - combining synergists in the resin (e.g. het acid resin),
  - adding epoxy-layered silicate nano-composites at the time of formulating the resin. The process is complicated and at present is expensive for the civil engineering industry, Hackman and Hollaway [77].

(c) To apply a passive fire protection system to treat the surface of the manufactured composite by using intumescent coating technology. These coatings incorporate an organic material which will char and evolve gases at a designed temperature so as to foam the developing char, Correia et al. [49] and by Keller et al. [122].

These approaches are discussed further in Section 5.1.3 concerned with fire in FRP composites.

2.3.5. Ultraviolet light (UV)

The ultraviolet light from the radiation of the sun is strong enough to cleave the covalent bonds in organic polymers, causing yellowing and embrittlement. All polymers are susceptible by varying degrees to the degradation by UV light. For a high degree of UV resistance, UV stabilisers are incorporated into the polymer during manufacture. Designers should seek advice from the manufacturer of the specific materials regarding their UV resistance to ascertain whether the UV stability is an important performance parameter.

2.4. The long-term in-service properties of the thermosetting polymers

2.4.1. Introduction

As briefly mentioned in Section 1 the polymer serves a number of functions besides being the binder to hold the fibres together in their required positions. It provides environmental and damage protection to the fibres and toughness to the composite. In addition, the polymer has important in-service properties, the density, the bonding and the degree of cross-linking of the molecular structure of the polymer are all function of its short term strength; it also depend upon the type of loading applied to that polymer. The long-term stability of the polymer will be dependent upon its durability in the environment into which it is placed. The stiffness of the polymer is a function of its degree of cure which in turn is a function of the degree of cross-linking of the three-dimensional network of polymer chains; however, the stiffness and strength of the polymer are not critical in terms of the composite as the fibres are the stiffening and strength component of the composite. What is important is the ability of the material under load to resist the particular civil engineering environment into which it is placed.

It should be mentioned that all materials will degrade over time and polymers (and composites) are more resistant to degradation than many of their competitors.

2.4.2. Durability

Karbhari et al. [116] noted that although the term 'durability' is widely used, its meaning and implications are often ambiguous. The authors defined the durability of a material or structure as *its ability to resist cracking, oxidation, chemical degradation, delamination, wear, and/or the effects or foreign objects damage for a specified period of time, under the appropriate load conditions, under specified environmental conditions*. The reduction in the material properties of the polymer (or composite) by the slow and irreversible variation of the structure of the polymer, morphology and/or composition as defined (above) is a chemical change in the polymer leading to its aging. The instability of the material during in-service use or its interaction with the environment into which it is placed is one of the causes of this change. One of the major concerns of the material is the ingress of moisture and aqueous solutions or the contact with an alkaline environment.

One of the main problems in undertaking detailed analysis of any durability property is the length of time involved in gathering the relevant information. There are many different polymers that are available to the civil engineer and some of these have been modified by chemists over the years to improve a particular physical and in-service property. In addition, additives are on occasions incorporated into polymers at the time of manufacture to enhance particular properties. Each time these polymers are changed/modified the durability will be affected.

The durability of a polymer is a function of the aggressive environments into which the polymer is placed.

These environments will now be discussed.

2.4.2.1. Polymer permeability/barrier property. Moisture will diffuse into all organic polymers leading to changes in their mechanical, chemical and thermophysical characteristics. The absorption of the moisture will cause mechanisms to be set up such as plasticization, saponification or hydrolysis that will cause both reversible and irreversible changes in the structure of the polymer. A high degree of cross-linking of the polymer leads to a decrease in its permeability with a consequent decrease in the diffusion process, thus it is necessary to fully cure the polymer. A successful method to decrease the diffusion for civil engineering polymers is to apply an additive to the matrix polymer at the time of manufacture. Silanes (organofunctional trialkoxysilanes) or organotitanates are two agents which have been used as a barrier against moisture ingress, van Ooij et al. [220]. Furthermore, epoxy-layered silicate nanocomposites introduced into the polymer at the time of manufacture has the potential to lower its permeability, thus improving its barrier properties and its mechanical strengths, Hackman and Hollaway [77]. Thus by improving the barrier property, a reduction of the ingress of moisture, aqueous and salt solutions is achieved. However, the utilisation of nano-composites is expensive for the construction industry and currently it would be used only under very special circumstances in construction.

2.4.2.2. Corrosion resistance. The resistance of thermosetting polymers to chemical attack depends upon its chemical composition and the bonding in its monomer. These polymers can degrade by several mechanisms, but degradation may be divided into two main categories, (i) physical and (ii) chemical.

- Physical corrosion is the interaction of a thermosetting polymer with its environment causing an alteration in its properties but no chemical reaction occurs.
- Chemical corrosion is when the bonds in the polymer are broken by a chemical reaction with the environment in which the polymer is situated. During this process the polymer may become embrittled, softened, charred, delaminated, discoloured or blistered; these are usually a non-reversible reactions. A correct curing procedure of the polymer is important to reduce these degrading effects.

Thermosetting polymers have a poor resistance to concentrated sulphuric and nitric acids. Furthermore, the attack of aqueous solutions occurs through hydrolysis in which moisture degrades the bonds of the polymer molecules. Polymers with high crystallinity/density or a high degree of cross-linking will generally have low permeability, thus gasses and other small particles will not readily permeate through it. Haque et al. [78], Liu et al. [143], Hackman and Hollaway [77] have shown that the ingress of moisture will permeate through polymers over time particularly if the polymer (and therefore the composite) is permanently immersed in water or salt solution or is exposed to de-icing salt solutions.

There are two-ways of measuring durability of polymers:

- Long-term testing in the natural environment.
- Accelerated test procedures.

These two methods of testing for durability are described in Hollaway [98], which discusses the durability of FRP composites.

3. Mechanical properties of the thermosetting polymer

As mentioned in Section 1 the mechanical properties of the component parts of the FRP composite will not be discussed in full. The main headings of the mechanical properties of polymers which must be taken into account when designing FRP composites for construction are:

- *Ultimate tensile strength*. Information may be obtained from Hollaway and Head [90]. Some typical tensile properties are given in Appendix A.

- *Ultimate compressive strength.* The compressive strength of thermosetting polymers is usually 1.5–4.0 times higher than in tension. Such a difference can be caused by the presence of various defects in the material, including micro cracks, whose influence is more pronounced in tension. In compression such cracks can be closed, which creates preconditions for achieving the yield point of the material.
- *Creep characteristics of polymers.* Information may be obtained from references Hollaway [98], BS 4618-5.3: 1972, Hollaway [93].
  (a) The time temperature superposition principle (TTSP) [7]
  (b) The time, applied stress superposition principle (TSSP), [37],
  (c) Further descriptions of the TTSP and TSSP may be found in Hollaway [93].
  (d) Cheng and Yang [38] have developed the (TTSP) further by introducing a matched theoretical calculated curve from a supposed model of transition kinetics in which only time is involved as the independent variable.
- *The uniaxial compressive strength* of fibre-reinforced polymer FRP composites is a very complex issue which is still not fully understood. Although FRP composites characteristically possess excellent ultimate and fatigue strength when loaded in tension in the fibre direction, compressive properties are not good. Unlike tensile properties which are fibre dominated, compressive properties are dependent upon other factors such as matrix modulus and strength, fibre/matrix interfacial bond strength, and fibre misalignment. Discussions on this topic are given in Hull and Clyne [107], ASTM D6641/D6641 M-09 [12].
- *Stiffness of polymers*. Discussions on this topic are given in Holliday and White [100].

Subramaniyan et al. [203], have shown that by the addition of nanoclays to the polymer the compressive strengths of GFRP composites increase.

4. The important physical and in-service of fibres

4.1. Introduction

Fibres can be formed from a wide range of amorphous and crystalline materials but in the construction industry the three fibres which are generally used in structural systems are the glass fibre (the E-glass fibre, the S-glass fibre and the Z-glass fibre), the aramid fibre (the aromatic polyamides, Kevlar 49 fibre) and the carbon fibre (the ultra high-modulus fibre, the high-modulus fibre and the high-strength fibre). The fibres may be used separately or as a hybrid of two or three different fibres. The various types and mechanical properties of glass fibre, the three types of carbon fibre, and the aramid fibre are discussed in Hollaway and Head [90] and Hollaway [97]. The basic manufacturing techniques for the ultra-high-modulus, the high-modulus and the high-strength carbon fibres are the same but the heat treatment temperature will be greater the higher the modulus of the fibres, thus, at the highest heat treatment temperature, about 2400° C. (for civil engineering fibres). The precursor polyacrylonitrile fibres are used for the production of high-modulus fibres (construction industry) or the production of high-modulus or ultra-high-modulus (aerospace industry). Pitch fibres which are derived from petroleum, asphalt, coal tar and PVC, the carbon yield is high but the uniformity of the fibre cross-sections is not constant from batch to batch; these fibres are used for the ultra-high-modulus carbon fibres (construction industry), Philips [178], Hollaway [87,98]. The definitions of fibres used here are the European ones, the US and many countries in the Far East use the normal and high-modulus terms when discussing high-modulus and ultra-high-modulus carbon fibres, respectively.

A typical sequence of operation used to form carbon fibres from polyacrylonitrile (PAN) precursor include various processes, these are:

- Stabilisation process in an air oven (oxygen is absorbed) to achieve dimensional stability. Temperature 200–300° C.
- Carbonation process is performed in and inert atmosphere (Carbon crystallites formed). Temperature > 800° C.
- Graphitisation process. (Fibres highly orientated.) Temperature > 1200° C.
- Surface treatment.
- Fibre winding process.

The three carbon fibres have very different strength and stiffness values, the ultra-high-modulus carbon fibre has a typical stiffness value up to 400 GPa (for civil engineering, this value can be increased under higher heat treatment) but a relatively low tensile strength value of 1800 MPa and therefore will have a low strain to failure value whereas the high-modulus carbon fibre has a typical stiffness value of 240 GPa and strength value of 4000 MPa and therefore a relatively high strain to failure value. The high-strength carbon fibre has typical tensile strength values of 4400 MPa and modulus values of 200 GPa.

The in-service properties of the three main civil engineering fibres are similar and will be discussed in the following section.

4.2. In-service properties of civil engineering fibres

4.2.1. Creep

The creep characteristics of glass, aramid and carbon fibres are very small and are not generally considered in the design of polymer composite components for civil engineering.

4.2.2. Durability

All glass fibres are very susceptible to alkaline environments, which is primarily due to the presence of silica in the glass fibres. These conclusions have been made when glass fibres (and therefore GFRP composites) are immersed into concentrated alkaline solutions. Mufti et al. [172], however, have shown from field surveys that the attack is minimal under civil engineering environments. There are, nevertheless, glass fibres on the market that are more resistant to this environment and are used to increase the durability of composites. Advantex, and ARcoteXTM are glass fibres which increase the durability of GFRP composites; the former is manufactured by Ownes Corning, and the latter by Saint-Gobain Vetrotex. Carbon fibres do not absorb liquids and are subsequently resistant to all forms of alkali or solvents ingress, Ceroni et al. [36]. Aramid fibres have been reported to suffer some reduction in tensile strength when exposed to an alkaline environment, Uomoto and Nishimura [216].

4.2.3. Effects of hydrolysis

Most glass fibres have limited solubility in water but they are very dependent upon the pH value of the liquid. Jones and Chandler [112] have stated that glass fibres are susceptible to chemical corrosion when exposed to strong acidic environments (pH value considerably lower than 7) and it is well known that they are also susceptible to high alkaline environments (pH values considerably higher than 7) in which they are severely degraded due to a combination of mechanisms ranging from pitting, hydroxylation, hydrolysis, and leaching. Chloride ions will also attack and dissolve the surface of the E-glass fibre. Moisture is readily adsorbed and can exacerbate microscopic cracks and surface defects in the fibre and thus reduce the tensile strength of the glass fibre. The glass fibres have high ratios of surface area to weight but the increased surface makes them much more susceptible to chemical attack.

4.3. The mechanical properties of the fibres

The tensile strength and stiffness of the reinforcing fibre are two of the most important mechanical characteristics. These properties have been discussed in Hollaway [98].

5. The civil engineering composite materials

5.1. Introduction

The short term in-service and physical properties of the polymer/fibre composite are governed by:

- The basic physical and in-service properties of the polymer and the fibre; the former is the most important component. These have been briefly discussed in and Sections 2 and 4.
- The surface interaction of fibre and resin (the 'interface').
- The relative proportions of the polymer and fibre, (fibre volume fraction – the fibres must be well wetted by the resin for this latter material to be fully effective, this becomes increasingly more difficult the higher this ratio).
- The method of manufacture of the composite.

The long-term durability of the composite will depend upon:

- The type of loading which the composite has to resist and the environment into which it is placed; there are two areas to be satisfied:
  1. The in-service properties – these are largely dependent upon the matrix material. (These properties of the matrix material have been discussed in Section 2.4.)
  2. The mechanical property of the composite, which are dependent upon the fibre and the interface between the two component parts.

5.1.1. The surface interaction of fibre and resin

The mechanical performance of a composite material is highly dependent upon the quality of the fibre–matrix interface. This region is an anisotropic transition region which is required to provide chemical and physical bonding between the fibre and the polymer. The primary aim of a fibre reinforced matrix composite material is to provide an average behaviour of the composite from the properties of the components which must act compositely for the material to be efficient. It is well known that the application of a coupling agent to, say, a glass fibre surface will improve fibre–matrix adhesion in that composite but in addition, and to a greater degree, it is the mixing of the processing additives; this contribution to composite properties is not well understood. The interfacial region of the composite will therefore be affected not only by the composition of the coating, but also by its distribution on the glass fibre surface and in the composite matrix.

5.1.2. The method of manufacture of the composite

There are three basic methods for the manufacture of composites:

- Manual methods – wet lay-up process:
  (i) The REPLARK method
  (ii) The Dupont method
  (iii) The Tonen Forca method.
- Semi-automatic methods
  (i) The hot-melt factory-made pre-impregnated fibre (pre-preg).
- Automatic methods
  (i) The pultrusion technique
  (ii) The filament winding method.
  (iii) The Resin Transfer Moulding process.

These techniques have been discussed in Hollaway and Head [90]. The automated fabrication methods have a high degree of production control, composite compaction and curing compared to the manual fabricated techniques and therefore the former technique will have higher values of the in-service properties (and strength and stiffness values) compared to those of the latter methods and therefore a more robust resistance to hostile environments.

Quality control and quality assurance are important aspects of composites during manufacture as both are dependent upon the performance characteristics and the overall integrity and durability of the composite formed. These aspects are particularly important for civil structures as they are required to withstand harsh and varying environmental exposure for long periods of time, (e.g. many bridges are designed for some 100 years). Clarke [42] has discussed the importance for a regular inspection regime on all structures, new or upgraded; these inspections are normally undertaken for bridges but few buildings are regularly checked. The inspections on the latter are carried out only when there is a change of ownership. Clarke recommends that all building owners instigate a regular inspection regime.

A further consideration is that many civil structures are fabricated on site with the likelihood of the technique of bonding being used; the bonding area is likely to be the weakest link, Mirmiran et al. [163]. Achieving a reliable standard of product requires good quality assurance procedures; this will be easier under factory controlled conditions compared with the more difficult site controlled conditions. Clarke [42] has discussed quality control and assurance.

5.1.3. In-service properties of FRP

5.1.3.1. Fire resistance. The property of the polymer in a fire has been dealt with in Sections 2–4. It is the polymer which protects the fibre and hence is the main component of the degradation of the composite in a fire; the following discussion will involve many problems met in Section 2.3.4.

Problems associated with the fire resistance of FRP composites are seen by many civil engineers as the single most critical technical barrier to the widespread use of structural engineering applications in the civil infrastructure. Until recently, only a few research groups worldwide had worked on this problem as, due to its complexity, it is not as amenable to the simpler types of modelling approach, Mouritz and Gibson [157]. Furthermore, the experimental testing regime for fire properties has not progressed as rapidly as that for the testing of mechanical properties of composites.

When FRP composite materials are exposed to high temperatures (300–500° C) the polymer matrix will decompose and will release heat and toxic volatiles. When heated to lower temperatures in the region of 100–200° C), FRP composites will soften, creep and distort, and this degradation of the mechanical properties often leads to buckling failure mechanisms of load-bearing composite structures [168]. This degradation will seriously compromise the structural properties of the FRP material which may lead to failures that could jeopardise the building, the building occupants as well as the fire fighting crews. Nevertheless, numerous research studies have shown FRP materials to be suitable for a variety of civil engineering applications but there is a limited amount of information regarding their behaviour in fire, Harries et al. [81] and Karbhari et al. [116], consequently, there is a barrier to its use in many building applications. For instance, Keller et al. [122] has stated that in Switzerland residential buildings with more than three floors a 90 min fire endurance is required, consequently, the use of FRP in buildings and parking garages has so far been limited. Externally-bonded FRP strengthened concrete structures are currently required to meet the minimum strength requirements of the un-strengthened concrete structure in fire thus any strength contribution from the FRP is ignored in a fire situation, [3]; this requirement is generally followed in European countries. Chowdbury et al. [41], Chowdbury et al. [39], Chowdbury [40], Bisby et al. [28] have shown that with appropriate insulation to the exterior of the FRP strengthening system, concrete structures strengthened with FRP materials can achieve an endurance ratings of greater than 4 h. However, after exposure to a severe fire, well-insulated RC members are able to retain most, if not all, of their original un-strengthened flexural capacity, particularly if the temperature of the compression concrete and reinforcing steel are maintained below 200° C and 593° C, respectively, Chowdbury et al. [41].

5.1.3.2. Alkaline and ingress of liquid into FRP composites. Matthews and Rawlings [149] found that the mechanical properties of FRP composite materials exposed to moisture penetration depended upon the sensitivity of the composite to matrix properties, measured as the fibre to polymer tensile strength ratio. Carbon-fibre-reinforced polymer (CFRP) composites have a high-strength ratio, which makes them usually unaffected by moisture absorption. In contrast, glass fibre-reinforced polymer (GFRP) composites have a low-strength ratio, making them susceptible to moisture attack. Ceroni et al. [36] present a state-of-the-art of the durability of FRP rebars which highlights issues relating to the material properties and interaction mechanisms that influence the service life of RC elements.

5.1.3.3. Permeability. Within a FRP composite, the polymer matrix offers the fibre some protection from moisture attack. However, it is relatively inefficient especially at normal fibre volume fractions of 60–65% where the average distance between the fibres is of the order of 2 μm or less. Methods to improve the permeability of FRP composites are:

- To apply a thin (few mm) polymer coating (gel-coat) to the outer surface of GFRP structures as a moisture barrier. However, this layer does not offer sufficient protection against moisture intrusion.
- The successful use of GFRP composites in wet environments has been largely due to the development of coupling agents that are applied directly onto the fibre at the time of manufacture. As with the protection of polymers against moisture ingress (Section 2.4.2.1), silanes (organofunctional trialkoxysilanes) or organotitanates are two agents which have been used.

5.1.3.4. Durability of FRP composites. As stated in Section 2.4.2 the durability of a FRP structural composite depends intrinsically upon the components of the composite, but in particular on the polymer and is a function of the environments into which it is placed. Therefore the composite durability is related to various factors, such as the properties of the resin, fibre and interfacial characteristics, fabrication process and the environment. The durability of FRP composites has been defined in (Karbhari et al. [116] for thermosetting polymers in Section 2.4.2.

Thus, the results of accelerated and other tests carried out in the laboratory must represent those in the field and an appropriate test programme taking into account the actual usage should be developed, on a case-by-case basis. [114,224,217] undertook tests on fibres and FRP rods to study the alkaline resistance, UV resistance, freeze thaw resistance, high temperature resistance, fire resistance and static fatigue fracture. The results showed:

- Carbon fibres and FRP rods had good durability characteristics.
- Aramid fibres and FRP rods had good durability properties except under static fatigue, UV radiation and acidic environment.
- Glass fibres had poor durability characteristics as far as their alkaline resistance is concerned, although they had satisfactory characteristics in an acidic and freeze thaw environment. FRP materials in general showed poor performance at high temperatures and therefore their use should be avoided when fire resistance is required.
- There is a need to limit the tensile load depending on the duration of the load in cases where the FRP are used as internal reinforcement.

There are two methods which are be used to obtain information on the durability of a material. These are field and accelerated tests, details of these have been given in Hollaway [92,98].

5.1.4. The type of loading which the civil infrastructure composite has to resist

- Tensile properties.
- Compressive properties.
- Impact resistance.
- Fatigue loading.
- Blast loading.
- Creep loading

The above mechanical properties have been discussed in Hollaway [98]. An overview of the blast loading and blast effects on structures is given Ngo et al. [174]. The impact of initially stressed composite laminates has been discussed by Sun and Chen [204], Kim [126]. Longinow et al. [142] has discussed research needed to resist terrorists attack.

6. Part B: the utilisation of FRP composites in the civil infrastructure

Part B will illustrate the types of structural systems that have been developed as a result of the unique physical and in-service properties of FRP composite materials; their present and future development have been and will be influenced by these properties. Fig. 1 illustrates the development of the fibre matrix composite from the early 1970s into the 21st century.

The building industry was the forerunner for the use of composites in the construction industry, with the introduction of Radomes during the Second World War. The material used for these structures was GFRP (known then as GRP) as it minimally attenuates the radio waves passing through them; therefore, it had the physical property of being transparent to radar or radio waves. During the 1950s and into the 1960s the FRP material in building and in construction had a very chequered existence with inexperienced fabricators, generally consisting of one or two operatives, manufacturing the composite without understanding the fundamentals of fabrication or the importance of the correct procedure for curing the material. By the 1970s consulting architects and civil engineers commenced to consider FRP composites as a building material and to design composite building structures. The larger fabricating firms, which already had experience of manufacturing FRP composite units for other industries, entered the building industry with the fabrication of semi-load bearing and infill panels for houses and larger constructions. The main large building examples of these systems in the UK are the class-room structure at Fulwood, Lancashire, the Mondial House, London and the Amex House in Brighton. The fabrication of the panels for these buildings was by

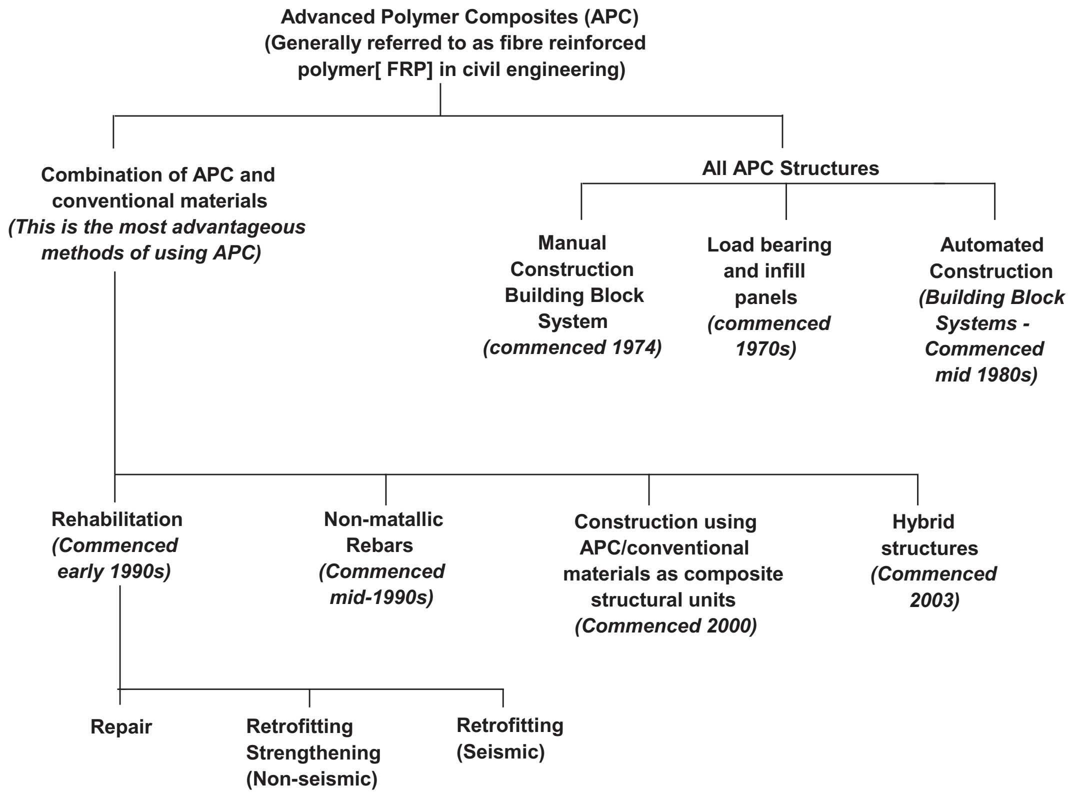

Fig. 1. The development of the fibre matrix composite from the early 1970 into the 21st century.

the hand lay-up method (which has since been updated and is known as the wet lay-up method), the first building used a glass fibre chopped strand mat and the latter two used a combination of glass fibre/polyester chopped strand mat and a 0/90 fibre array mat; the polymer used in all cases was the polyester. There are many other examples where GFRP composites were used for semi-load bearing infill panels and where GFRP panel 'buckets' were used in conjunction with a steel double layer skeletal structure to construct roof systems such as the Covent Garden Market at Nine Elms, London. The shapes of the above structures were largely of a folded plate construction in order to add stiffness to the overall structure as the stiffness of GFRP composites were/are low.

The class-room building, in Lancashire was/is 'all composite' FRP building in the form of a geometrically modified icoshedron and was manufactured from 35 independent self-supported tetrahedral panels of chopped strand glass fibre reinforced polyester composite; the geometrical shape was of folded plate construction likewise to provide stiffness to the structure. Twenty-eight panels have a solid single skin GFRP composite skin and in five of these panels circular apertures were constructed to contain ventilation fans. In the remaining seven panels non-opening triangular windows were inserted. The wet lay-up manufacturing method was utilised to manufacture the E-glass fibre/polyester composite skins. The inside of the panels have a 50 mm thick integral skin phenolic foam core acting as a non-load bearing fire protection lining to the GFRP composite skins; the foam core also maintains a constant temperature in the class-room, (see Section 2.3.3 for the thermal properties of foam polymers). The geometric icoshedron structure is separated from the concrete base by a timber hardwood ring. The FRP panels were fabricated onto a mould lining of perspex with an appropriate profile to give a fluted finish to the flat surfaces of the panels. The edges of the panels were specially shaped to provide a flanged joint which formed the connection with adjacent panels. Sandwiched between two adjacent flanges is a shaped hardwood batten, this provides the correct geometric angle between the panels; the whole is bolted together using galvanised steel bolts placed at 450 mm intervals. The external joint surfaces between the adjacent panels were sealed with polysulphide mastic. The glass windows were fixed in position on site by means of neoprene gaskets. The architect to Lancashire County Council conceived the idea of the GFRP class-room and developed a complete school manufactured from GFRP; this latter idea never developed further than the 'drawing board'. The single class-room was build and continues to be used by the pupils at the Fulwood, Lancashire school.

In 1974, Mondial House situated on the banks of the Thames between Cannon Street station and London Bridge was one of the most prestigious and ambitious building projects both in terms of size and complexity; it was designed as a skeletal RC beam and column structure and clad above the upper ground floor level. The semi-loading cladding panels were contact moulded using Scott Bader Crystic 356PA 'Class O' fire resistant polyester laminating resin and isophthalic polyester gel-coat 65PA for weather resistance and durability; glass fibre was used as the reinforcing component. In 2007 as telephone exchanges no longer require vast amounts of space and, in addition, to allow for redevelopment of that area it was vacated by the Post Office and was demolished. It was, when erected in 1974, the largest exchange in Europe; it was built as a bomb proof structure at the height of the cold war. The outer skin of the GFRP panel included a gel-coat which used isophthalic resin, pigmented white, with an ultraviolet stabilizer backed up with a glass fibre-reinforced polymer laminate; the latter used a 3oz per square foot chopped strand mat and a self-extinguishing laminating resin reinforced with 9oz per square foot glass fibre chopped strand mat reinforcement. Some degree of rigidity was obtained from a core material of rigid polyurethane foam bonded to the outer skin and covered on the back with a further glass reinforced laminate; this construction also provided thermal insulation. Further strength and rigidity was obtained by the use of light-weight top hat section beams, manufactured as thin formers and incorporated and over-laminated into the moulding as manufacture proceeded. The effect of the beams was transferred to the front of the panel by means of glass fibre reinforced ties or bridges formed between the polyurethane foam at the base of each beam. The face of the beam was reeded on the vertical surfaces in order to mask any minor undulations and to provide channels off which the water ran and thereby cleaned the surface. The reeding also gave the effect of a matt panel without reducing the high surface white finish. The structure was visually inspected in 1994 by a member from Scott Bader and one from University of Surrey and the degradation was found to be minimal.

The American Express Building in Brighton was completed in 1977 and was build as a composite construction with a skeletal load bearing structure with RC columns, major RC beams and panels manufactured from GFRP composites. These latter were used as beams spanning between 7.2 m and 12.5 m. The GFRP beams supported 2 m high laminated glazing system and the roof structure. This application is more significant from a structural engineering point of view as it involves a semi-structural application of GFRP composites. A description of the structure is given by Berry [26] and Roach [188].

Architectural embellishment rather than structural performance and durability motivated the early applications of GFRP composite materials. By the mid-1980s there was a desire by engineers to use FRP composites as a structural material and this was driven by the need for durable, high-strength and high stiffness materials that could replace the more conventional civil engineering materials in aggressive and hostile environments that are sometimes encountered in civil engineering applications. Thus consulting civil/structural engineers investigated the possibility of using automated manufacturing methods for the manufacture of structural components for 'all composite' structures; the main one chosen was the pultrusion technique. At this time structural unit building blocks were being considered and Maunsell Structural Plastics, (Faber Maunsell now AECOM), Beckenham, Kent designed and developed ACCS Plank known as the Manusell Plank. The ACCS construction consisted/consists of a number of interlocking fibre-reinforced polymer composite Maunsell Plank units which can be assembled into a large range of different high-performance structures for use in the construction industry; the panels were connected to each other and to the connectors by bonding and GFRP toggles were used to maintained the parts together whilst the adhesive polymerised. The details of the Maunsell Plank are shown in Hollaway and Head [90]. The system was initially manufactured by the pultrusion technique using isophthalic polyester resin and uni-directional, bi-directional and chopped strand mat glass fibre reinforced for the main structural members. The production and material content of the ACCS plank are optimized to provide highly durable and versatile composites and, in addition, structures can be formed quickly from a small number of standard components. The ACCS system is now marketed by Strongwell Corporation, USA as Composolite.

Kendall [123] has given a review of the type of FRP building structures for the future.

From the mid to late 1980s the use of FRP composite materials commenced to expand and the following section will discuss some of these systems; Canning et al. [32] has discussed the use of advanced composites in the civil infrastructure.

6.1. The All FRP composite bridge structures

The first pedestrian FRP bridge was built in Tel Aviv, Israel in 1975, [83]. Since then, others have been constructed in Asia, Europe, and North America. Many innovative pedestrian bridges have been constructed using pultruded composite structural shapes and due to the light-weight materials and the ease in fabrication and installation many of these bridges have been able to be constructed in inaccessible and environmentally restrictive areas without having to employ heavy equipment. Some bridges were flown to the sites in one piece by helicopters; others were disassembled and transported by vehicles and assembled on site. The advancement in this application has resulted in the production of second generation pultruded shapes of hybrid glass and carbon FRP composites that will increase the stiffness modulus with very little additional cost. The recognition of providing high quality fibres at the most effective regions in a structural member's cross-section is a key innovation to the effective use of these high-performance materials.

The first cable stay, GFRP deck and pylons footbridge was conceived and developed in the UK and was erected at Aberfeldy, Scotland in 1992; this bridge crosses the river Tay in Scotland and joins two parts of the Aberfeldy golf course. The deck and pylons were constructed from interlocking ACCS Planks; the use of composite materials resulted in a light-weight structure, which could be erected without the aid of heavy machinery. The durability performance of this bridge over 16 years of service has been very satisfactory, Stratford [202].

The Bonds Mill Road Bridge, Gloucestershire, England, UK crosses the Stroudwater Navigation canal near Stonehouse, Gloucestershire, England. It is an electrically operated single bascule lift bridge and was completed in 1994; it is constructed from 10 ACCS units which form an integral 3D multi-cellular box structure 8.5 m span and 4.25 m wide and 0.8 m depth weighing 4.5 tonnes. The individual ACCS units are bonded together through square connectors containing recesses through which mechanical toggles are fastened to ensure the units do not slip during the bonding operation; they also act as mechanical fasteners. The bridge is able to support vehicles up to 44 tonnes weight. The box structure relies on cold cure adhesive bonding with an epoxy adhesives, Head [84].

The Wilcott Bridge, Shropshire, England was constructed in 2003 and is a 51.3 m single span suspension footbridge with a slightly cambered slender deck providing a footway 2 m wide. It was built in three units and spliced to fabricate the total length. The cross-section of the deck of the bridge consists of six GFRP 'Maunsel Plank' units; three units were placed in the top surface of the deck and three in the bottom. A description of the bridge is given in Faber Maunsell [67].

These three structures were innovative prestigious structures in their day and the two footbridge structures are ascetically pleasing but they were expensive to manufacture and fabricate.

Recently the UK Highways Agency has completed a new bridge over the M6 motorway consisting of steel primary beams and a pultruded FRP deck.

Strongwell, Bristol VA and Chatfield MN, USA, now hold the manufacturing licence for the plank and produce similar panel under the trade name of COMPOSOLITE®. Further information can be obtained on this Maunsell Plank system from Hollaway and Head [90] and Strongwell, Bristol, Virginia, USA. There have been some footbridges recently built that have used COMPOSOLITE®. For instance, the New Chamberlain Bridge, Bridgetown, Barbados, was completed in 2006. It spans the Careenage River, was designed by AECOM and was constructed from COMPOSOLITE®. This bridge replaced the steel horizontal swing bridge built in 1872. The composite bridge is composed of two 2.13 m (7 ft.) raised side walks with handrails and a 4 m (13 ft.) wide road in the centre which is for emergency vehicular traffic only; the lift bridge is of similar construction to that of the Bonds Mill bridge.

A bascule FRP composite footbridge of span 56 m was opened in May 2003. This double-cantilever bridge crosses the river Vesterelven in Fredrikstad, Norway; it is the largest moveable bridge in Scandinavia. One large hydraulic cylinder operates each of the 28 m long cantilevers to open and close. The cantilevers are built as a closed box girder with double curved outer surfaces and longitudinal and transverse stiffeners. The approximate weight of a fully equipped unit is 20 tonne of which 9 tonne is the weight of the FRP composite materials. The bridge deck is of sandwich construction with CFRP laminates and a Balsa wood core; there are embedded heating cables for defrosting during winter. The deck is sufficiently strong to carry a car with up to 2.0 tonne axel load. The bottom flange of the girder is manufactured from single skin CFRP laminates (10–38 mm thick). The internal stiffeners are all sandwich constructions with CFRP and/or GFRP laminates and PVC core materials. All FRP composite material was manufactured by vacuum assisted resin infusion. A steel construction inside the thickest end of the girder is used to distribute the concentrated bearing loads. The client was Værste AS and Fredrikstad commune, the designer was Griff kommunikasjon AS and the construction was undertaken by Marine Composites AS, Arendal, Norway.

A footbridge installed by Network Rail was erected over the main Penzance-Paddington railway line at St. Austell, Cornwall during the weekend of 21–22nd October 2007. It has a central span of 14 m and three sections each 6 m wide; the composite sections were manufactured by Pipex (Plymouth) and the consultant for this bridge was Parsons Brinkerhoff.

The 47 m span Halgavor suspension bridge which spans the A30 road near Bodmin, Cornwall is one of the longest curved composite structures in Europe; it was opened in July 2001. The bridge deck was designed to carry pedestrians, cyclists and horses is constructed of composite materials with bonded structural joints. The 4 m wide deck is supported from four cranked steel masts at the abutments using suspension cables along the length. The FRP deck was manufactured by Vosper Thornycroft using resin infusion with vinyl-ester resin and an ultraviolet (UV) resistant gel-coat. Polyester pultrusions were also used longitudinally to locate the deck. The composite solution provided easy installation and gave to the architects' freedom to design the required shapes, colours and lines.

The first Russian composite bridge made by vacuum infusion resulted in the development of a product line of arched bridges for rivers with span lengths between 15 m and 30 m and a life cycle span of 100 years. The use of the vacuum infusion technology provided reduced manufacture steps, avoided site assembling and thus decreased the cost of the structure. The first bridge was installed at the p. Vernadskogo subway station, Moscow in 2008, the structure was designed by light-weight structures BV, the Netherlands and installed by Applied Advanced Technology (ApATeCh), Company Ltd., Ushakov et al. [215]; the paper won the best innovative construction paper award from the American Society of Civil Engineers.

A bridge built of composite materials can be constructed and put into service in a relatively short time and at a competitive cost. Its light-weight materials and ease of construction provide large labour and traffic control cost savings to offset a higher first cost. Mosallam [167] wrote a state-of-the-Art review of composites for highway bridge applications.

Keller [121] has presented a review of all-composite bridge and building construction from 1997 to 2000.

7. The combination of FRP composites with other materials to form hybrid systems

7.1. Introduction

The unique properties of advanced polymer composites in the civil infrastructure suggest their suitability for integration in hybrid structural systems as well as the development of hybrid FRP materials themselves. Hybrid structures are those in which two (or more) dissimilar materials could structurally compliment each other. Hybrid systems range from open or closed stay-in-place formwork to hybrid structural systems, incorporating FRP and conventional construction materials. The combination can take many structural forms, these are:

- All FRP composite bridge decks and the bridge superstructure.
- An access to an existing structure for maintenance purposes and for aerodynamics of the structure – a bridge enclosure and aerodynamic fairings using FRP units.
- The rehabilitation of RC beams by the techniques of (i) external plate bonding (EPB) and (ii) Near Surface Mounted FRP rods.
- The rehabilitation of steel beams by the techniques of EPB.
- The retrofitting of RC columns by using uni-directional FRP composites.
- The FRP rebars used to reinforce concrete beams and slabs.
- The construction of a structural member to enable two or more materials to take advantage of their superior properties. For instance, combining FRP composites with concrete which is weak in tension but strong in compression whereas FRP composites in plate form are strong in tension but will buckle under low compressive loads. The combination of these two could take advantage of the dominant properties of both materials by joining the two materials to form a structural member.

The structural analysis and design of the above systems generally do not present many problems; in addition, there is evidence in the literature that provides substantial reasons to believe that, if appropriately analysed, designed and fabricated FRP composites can provide longer lifetime and lower maintenance costs than equivalent structures fabricated from conventional materials. However, there are areas within the physical and in-service properties of FRP composites that are sparse particularly on the durability of composites; this property will affect the long-term behaviour of the material. One of the problems with composite materials is in the general name of their component parts. For instance the matrix material is generally defined in the literature as the polymer but as stated earlier there are three polymers used in civil engineering construction (but there are many more which are used in engineering generally) each having different in-service properties. Furthermore, these matrices may have had additives applied at the time of their fabrication which will have affected their properties. Likewise with the fibres, there are many which have the same general name but within their family group will have differing properties. The lack of an easily accessible comprehensive data base on the mechanical and in-service properties of the groups of polymers and fibres and indeed also the lack of many codes of practice and specifications makes it difficult for the practicing civil engineer and designer to have the confidence to use FRP composites on a routine basis.

7.2. All FRP composite bridge decks

The bridge deck is the most vulnerable element in the bridge system because it is exposed to the direct actions of wheel loads, chemical attack, and temperature/moisture effects including freeze and thaw shrinkage and humidity.

The FRP bridge deck structures are typically made with vinylester polymer and E-glass fibre and are based on the pultruded manufacturing system; occasionally the deck is moulded. The FRP deck replacement can be manufactured in conjunction with the FRP superstructure replacement for the bridge; the deck is manufactured in a factory and the fabrication is undertaken on site, the wearing surface is then added.

The advantages of the FRP bridge deck are:

- *Light weight* – FRP bridge decks weigh about 10–20% of the structurally equivalent of a reinforced concrete deck. Consequently, using an FRP deck to replace a concrete deck reduces the dead load significantly. A lighter dead load can translate into savings throughout the structure and the foundations are reduced for new structures.
- *Corrosion resistance* – the corrosion of the reinforcing steel is the main cause of the premature deterioration of RC bridge decks. The use of road de-icing salts accelerates this corrosion. FRP composites have resistance against these corrosion forces. (see Sections 2.4.and 5.1)
- *Rapid installation time with minimum traffic disruption* – factory made FRP deck panels offers several advantages over cast-in-place concrete decks. These are:
  (i) Quality of the product can be closely monitored in the controlled factory environment.
  (ii) During manufacture the potential for inclement weather is eliminated.
  (iii) Once the superstructure is prepared, the fabricated deck structure can be installed quickly with light lifting cranes. [cf. cast-in-place RC deck site construction – erecting formwork, placing rebars, pouring and curing concrete and removing formwork.]
- *High-strengths* – stiffness drives the design of FRP decks, they have high safety factors; decks also have high ductility.
- *Lower life-cycle costs* – life cycle cost savings have been shown to more than offset the relatively high initial cost of the FRP materials compared to conventional materials; the service life of the FRP deck can be about three times greater than concrete decks. However, few public agencies select materials based on projected life-cycle costs, most materials are chosen on the experience and judgement of the engineer, agency preferences and industry standard practice, generally with a strong bias towards minimising initial construction costs.

The high-strength to low-weight ratio enables the bridge deck to carry the currently designed traffic loads with little or no upgrading of the superstructure. The dead load of the bridge deck is about 20% of the weight of an equivalent size of a RC deck and can be erected within 2 days. FRP composite bridge decks have been used in the United States since the mid-1990's; the span of these bridges are generally about 10–12 m.

The bridge market represents a major and largely untapped potential market for light-weight, corrosion resistant FRP composite materials. However, there are major barriers to the use of FRP bride decks, these are:

- *The cost of the FRP decks* – highway authorities responsible for construction and maintenance of the nation's bridges are under considerable pressure to maintain the significant number of substandard bridges all of which are competing for the limited amount of monies for such purposes. Under these conditions officials are compelled to maximise the number of bridges in serviceable condition at any given time and rarely have the latitude to consider the life-cycle costs advantages of initially more expansive materials. Consequently, any decision to use a more expensive material must be justified based on superior performance or specific project requirements.
- *Standard specifications* – specifications for the procurement and construction of FRP deck must be developed so that bridge owners can obtain the decks within their procurement process.

FRP bridge decks are required to meet the same design requirements as conventional bridge decks. Unless waived or modified by the bridge owner, typical design criteria are given in Daly and Duckett [50], AASHTO [1], AASHTO LRFD [2], BD 90/05 [23]. Most of the bridge decks which have been built use proprietary experimental systems and details, consequently, the lack of geometrical/material standardisation is a challenge to bridge engineers, who traditionally are accustomed to standard shapes, sizes and material properties.

Most of the deck systems are sealed and enclosed; they are inaccessible for field inspection. To ensure the composites' integrity, sophisticated non-destructive evaluation/testing (NDE/NDT) devices and fibre optic sensors have been incorporated into some of the composite deck systems to monitor the in-service condition of and the presence of moisture in the bridge deck. With time the effectiveness of the monitoring systems and the long-term service performance of composites can be ascertained.

In the UK the first bridge deck and superstructure replacement was demonstrated by the innovative ASSET Project, Luke et al. [146], conceived and developed by a European consortium led by Mouchel Consulting, West Byfleet, UK. This project culminating in 2002 in the construction of the West Mill Bridge, over the River Cole in Oxfordshire; the beam and deck structures were manufactured by the pultrusion technique, Zhang and Canning [225]. The span of the bridge is 10 m with a width of 6.8 m; the bridge carries two lanes of traffic and a footpath. The beams have uni-directional carbon-fibre-reinforced polymer composite flanges bonded to the GFRP beams to provide the required global flexural rigidity.

The first vehicle carrying FRP bridge deck in the UK to span over a railway, replaced the existing over-line bridge at Standen Hey, near Clitheroe, Lancashire; it has a span of 10 m, weighs 20 tonnes and was completed in March 2008, TGP [212]. This is the first of Network Rail's six trial sites in the country. The consulting firm, Tony Gee and Partners, was responsible for the design of the bridge deck which comprises of three layers of ASSET panel deck units which are made from E-glass fibres in the form of bi-axial mats within a UV resistant resin matrix.

Composite Advantage (CA) built April 2008 a new 'drop-in-place' GFRP composite pre-fabricated integral beams and deck bridge superstructure, 6.75 m long by 19.0 m wide (22 ft $\times$ 62 ft) in Hamilton County, Ohio, USA. No heavy lifting equipment was required and took 1 day to install, Composite Advantage Newsletter [44].

The UK Highways Agency in 2008 completed a new single carriageway road bridge over the M6 motorway. The superstructure comprises a novel pre-fabricated FRP deck spanning transversely over, and adhesively bonded to, two longitudinal steel plate girders. The FRP bridge deck constructed from ASSET construction was designed by Mouchel Group, Manchester, UK and provides general vehicular access to an equestrian centre; it was designed for unrestricted traffic loading, Canning [33].

The 27 m long bridge over the German B3 highway at Friedberg near Frankfurt comprises a superstructure of two steel beams with a multi-cellular GRP deck constructed of the ASSET pultruded profiles, [75].

FRP composites have a high tensile strength; however, in almost all of the demonstration bridge projects constructed to date, the design has been driven by the stiffness requirement rather than strength. Greater improvement and advancement of the composite deck systems will capitalize on its material strength. The key to successful application of the deck superstructure system is to optimize its geometric cross-section and to establish well-defined load paths.

The modular panel construction of bridge deck systems enables quick project delivery. A bridge built of composite materials can be constructed and put into service in a relatively short time and at a competitive cost. Its light-weight materials and ease of construction provide tremendous labour and traffic control cost savings to offset a higher first cost; to improve on the time element even further Lee and Hong [136] have developed an innovative profile of snap-fit connections for composite-deck bridges. An FRP deck could reduce the weight of conventional construction by 70–80%. This technology has demonstrated that a bridge structure can be replaced and put into service in a matter of hours rather than days or months; this is innovative technology put to good use.

7.2.1. Bridge enclosure and aerodynamic fairings using FRP units

Most bridges designed and built over the last 50 years do not have good access for inspection consequently the maintenance cost is increased by the erection of falsework. In 1982 the concept of a FRP 'Bridge Enclosure' was developed by Transport Research Laboratory (TRL, formerly TRRL), UK and Maunsell, Beckenham, UK, (now AECOM Europe), Head [85], to provide a solution to this problem. The design standard covering Bridge Enclosures was published by the Highway Agency, UK in 1996, BD67/96 [24], Enclosure of Bridges. The requirements for wind loading are covered by BD37/01 [25], UK, 'Enclosure of bridges'. The effects of traffic induced pressure waves, fire design and appropriate access and escape provision should also be covered. The first major example which utilised this technique in the UK was in 1988–1989 when the A19 Tees Viaduct at Middlesborough was fitted with the Maunsell 'caretaker' system, Constable [48].

Bridge enclosures facilitate bridge construction, inspection, maintenance, upgrading and operation with minimum traffic disruption. In addition, it provides access to the bridge bearings, drainage pipes and services, and provides corrosion protection, environmental protection, improved safety and convenient clear boundaries between responsible authorities. The floor of bridge enclosure is sealed onto the underside of the edge girder and once the enclosures are erected and sealed the rate of corrosion of uncoated steel in the protected environment within the enclosure is 2–10% of that of painted steel in the open, McKenzie [150,151].

Advantages of using enclosures and aerodynamic fairings are reductions in:

(i) The cost of maintenance due to fewer coats of steel bridges.
(ii) The concrete cover to the steel for RC members.
(iii) The wind loads on structure.
(iv) The costs of formwork and falsework.
(v) **T**he oscillation-inducing loadings on the bridge.

8. The rehabilitation of RC beams by the techniques of external plate bonding (EPB) and Near Surface Mounted (NSM) rods

8.1. Introduction

The deterioration of some civil engineering structural elements, in particular bridge systems, and the need to upgrade others to service requirements and capacities beyond those for which the systems were initially designed, has placed demands on owners and highway authorities to effect rapid renewal. The maintenance of these degraded structures has become one of the fast growing and important challenges confronting the engineer worldwide.

Throughout the industrialised world there are many bridges and building structures which are either structurally deficient or functionally obsolete. The definitions mainly refer to bridges and were defined by Hollaway [97]:

- A structurally deficient bridge is one whose components may have deteriorated or have been damaged, resulting in restrictions on its use.
- A functionally obsolete bridge refers to the geometrical characteristics of the bridge in terms of the load carrying capacity of it. For instance, a bridge which was designed some 40 years ago for lower load levels, traffic volume or under/over clearance and which now requires restrictions to be imposed on its use is functionally obsolete in spite of its good structural condition.

In the case of the civil infrastructure deficient structures can be split into two broad groups:

- Changes in the use of a structure, so that it needs to carry different loads from those originally specified.
- Degradation of a structure, so that it can not carry the loads for which it was originally intended.

Both of these broad classifications of structural deficiency can be addressed using FRP composites. The use of externally bonded plates and NSM CFRP systems to strengthen RC beams in flexure has been well researched, Meier [158], Hollaway and Leeming [89], De Lorenzis and Nanni [51,52], Teng et al. [207], Hassan and Rizkalla [82], Hollaway [94]. Each construction material has different properties, and will be required to perform different upgrading functions; consequently, structural deficiencies are discussed for each of the general FRP composite materials used in the civil infrastructure. Some structural deficiencies are common to any type of structure.

Civil infrastructure routinely has a serviceable life in excess of 100 years; in addition it is inevitable that the structure will be required to fulfil a role not envisaged in the original specification. These changes include:

- *Increased live load;* (i) increased traffic load on a bridge and (ii) change in use of a building resulting in greater imposed loads.
- *Increased dead and live load;* additional load on underground structures due to new construction above ground, for instance, widening a bridge to add an extra lane of traffic and additional allowable live load.
- *.Pre-stressed concrete beams;* strengthening measures may be required to prevent further loss of prestress.
- *Modern design practice;* changes in modern design technique requirements and updated current codes.
- *New loading requirements;* a structure not originally designed to carry blast or seismic loads.
- *Material deficiencies;* the degradation of the material of the structural system.

There are state-of-the-art papers, review papers and books on FRP composites on the civil infrastructure and on the rehabilitation of existing civil structures, Hollaway and Mays [88], Karbhari and Li [115], Bakis et al. [13], Motavalli and Czaderski [156], Hollaway and Teng [96],

8.2. Unstressed FRP soffit plate for flexural strengthening of concrete structures

The strengthening of concrete structures may be undertaken by externally-bonded FRP composites Hollaway [98] ; this technique is now routinely considered a viable alternative to the rather costly replacement of these structures. However, it should be mentioned that the designer should have knowledge of the in-service properties, particularly the durability of the composite material, which is to be used to rehabilitate the degraded member; Section 2.3 discusses the influence of temperature on polymers, and Section 2.4 discusses the long-term in-service properties of the thermosetting polymers. The first upgrading applications to RC beams used wet lay-up sheets or pre-cured pultruded plates bonded to the tension face of the beam with the fibre direction aligned to the beam axis. The high-strength-to-weight ratio and good corrosion resistance of FRP materials provides considerable advantages over that of steel for rehabilitation. The effectiveness of flexural strengthening of

RC beams with FRP is evident from the large database of experiments, reported by Smith and Teng some 8 years ago, Smith and Teng [198,199]. Furthermore, Porter and Harries [181] have discussed future directions for research into FRP composites in concrete construction.

The ductility of a flexural member generally decreases as a result of strengthening, especially if the controlling failure mode is de-bonding or FRP rupture. To guarantee adequate ductility of a strengthened cross-section, the strain level of the internal steel reinforcement at ultimate should considerably exceed the steel yield strain, as indicated by available design recommendations (e.g. Federation Internationale du Beton, fib Task Group 9.3 [71] and ACI 440.2R-08). ACI 440.2R-02 also suggests that the lower ductility should be compensated with a higher reserve of strength through the use of a lower overall strength reduction factor.

Under service loads, the effectiveness of a non pre-stressed FRP system is usually limited, but a notable increase in the ultimate moment of the cross-section can be obtained. The analysis of strengthened members at the ultimate limit state may follow well-established procedures valid for RC members, with the exceptions that any contribution of the FRP must be properly accounted for, and the issue of bond between FRP and concrete must be carefully designed and executed, Teng et al. [208].

FRP composites have also been used to rehabilitate pre-stressed concrete (PC) bridge members. PC members are susceptible to steel strand fatigue and may require strengthening to prevent further loss of prestress, Hollaway and Leeming [89], Hassan and Rizkalla [82]. Takács and Kanstad [205] showed that pre-stressed concrete girders could be strengthened with externally bonded CFRP composite plates to increase their ultimate flexural capacity. Reed and Peterman [185] showed that both flexural and shear capacities of a 30 year old damaged pre-stressed concrete girders could be substantially increased with externally bonded CFRP composite sheets. They used CFRP U-wraps as shear reinforcement along the length of the girder to delay de-bonding failure.

Governments and engineering associations worldwide are cooperating to standardise workable international design parameters, and the composites industry is building up critical alliances with the civil engineering community and associations. A notable example is the American Concrete Institute (which in 2008 published three new guides for concrete reinforcement: ACI 440-2R-08: guide for the design and construction of externally-bonded FRP systems: ACI 440.5-08: specification for construction with fibre-reinforced polymer (FRP) reinforcing bars: and 440.6-08: specification for carbon and glass Fiber-Reinforced Polymer (FRP) Bar Materials for Concrete reinforcement.

8.3. Preparation of adherent surfaces

Prior to the rehabilitation or retrofitting of RC and PC structures their surfaces to be bonded must be prepared, likewise, the surface of. the FRP composite. It is necessary to pre-treat the surfaces of the adherents to enable the required surface properties to be achieved. The concrete substrate is prepared by:

- grit blasting, and in the UK 'Turbobead' grade 7 angular chilled iron grit, Guyson [76] of nominal 0.18 mm particle size is generally used. The surface is then solvent degreased. This operation is important, because it removes contaminants, which inhibit the formation of the chemical bonds, Kinloch [127].

The FRP surface is prepared by either:

- *The abrasion method followed by solvent cleaning.* Abrasion removes weak surface layers and contamination and increases the apparent surface energy and the rate of spreading of the adhesive but it is important not to abrade down to fibre exposure. Although the degree of abrasion prior to bonding is known to affect subsequent bond strength and durability, the strength of bonded FRP joints depends on the roughness of the surfaces and the level of contaminants present.
- *The peel-ply method.* Peel-ply composites are adapted for the manufacture of multi-layer laminates built from glass and carbon fibre pre-pregs. Peel-ply layers can also be applied to pultrusion composites; they are attached to the pultruded units during the manufacturing procedure. A peel-ply is a layer of nylon or polyester fabric incorporated onto the surface of the composite during manufacture. The peel-ply is stripped from the pultruded surface immediately prior to bonding to the adherent to provide a clean, textured surface to the composite unit. Most peel-plies are coated with a release agent to ensure that their removal does not damage the underlying plies. Hollaway and Leeming [89], recommended the use of the peel-ply method particularly when long-span beams (e.g. 18 m span beams) are to be upgraded using strips of CFRP composite manufactured by the pultrusion technique.

Further information on the technique and analysis of rehabilitating FRP composites to reinforced concrete may be obtained from Hollaway and Leeming [89], Lam and Teng [132], Teng et al. [207], Lui et al. [145], Teng et al. [209], Anania et al. [9], Pešić and Pilakoutas [177], Lu et al. [144], Hollaway and Teng [96], Bogas and Gomes [29].

8.4. Failure areas of an upgraded RC beam

There are nine failure areas of a RC beam upgraded with an unstressed FRP plate. The following description has been developed from Hollaway and Leeming [89], where the various modes of failure have been discussed.

- For an un-strengthened and over-reinforced RC beam, the flexural failure occurs as a concrete compression failure at the top flange (mode 1).
- For an un-strengthened and under-reinforced RC beam, the initial failure occurs at yield of the steel tensile reinforcement (mode 2), with an increasing deflection but without any additional load carrying capacity, the beam fails in concrete compression in the top flange, (mode 1), due to excessive deflection.
- For an un-strengthened and under-reinforced beam and if the beam remains under-reinforced when strengthened with an FRP plate, the failure mode could be a tensile rupture of the laminate, (mode 3).
- For a beam over-reinforced after plate bonding, flexural failure occurs as a concrete compression failure in the top flange (mode 1). Yielding of the steel reinforcement is likely to occur before either the concrete or the CFRP plate fails and whilst this may contribute to the ultimate failure of the beam it is not the prime cause of failure. At the termination of the plate (plate free end) there are high normal stresses to the plate, these will cause the plate to peel off towards the centre of the beam; this is known as end anchorage peel, (mode 6 – anchorage peel/shear in cover zone)and (mode 7 – peel failure)
- For upgraded beams there is also a peel failure mode at a shear crack, (modes 4 – shear failure), (mode 5 – peel due to vertical movement at the shear crack) and (mode 8 – adhesive failure at concrete/adhesive interface), where there is a possible complex mechanism of de-bonding due to strain redistribution in the plate at the crack and/or the formation of a step in the soffit of the beam thus causing shear peel. The delamination can then propagate towards the end of the plate. Whether modes 5 or 8 occur depends upon the structure of the shear reinforcement in the un-strengthened beam.

There are a number of other possible but unlikely modes of failure which have been identified in the literature such as delamination of the composite plate or of the area within the glue line but these have not generally been experienced; the strength of these materials is higher than that of concrete and the failures will only happen if the installation has been poorly performed or there is a defect in the manufacture of the plate.

There are several design guides for the design calculations for retrofitting of FRP composites to reinforced concrete structures, used throughout the world, these have been given in the Appendix B to this paper.

8.5. Shear strengthening of RC structures

When a RC beam is deficient in shear or when its shear capacity is less than the flexural capacity after flexural strengthening, shear strengthening must be undertaken. FRP can be effective in increasing the shear capacity of RC beams and a number of different design approaches have been proposed Fanning et al. [70], Teng et al. [207], Barnes and Mays [21]. Efficient design requires the principal fibre direction to be parallel to that of the maximum principal tensile stresses, which generally is at approximately 45° to the member axis. However, for practical reasons it is usually preferred to attach the external FRP reinforcement with the principal fibre direction perpendicular to the member axis.

Different strengthening patterns can be used, both along the axis of the beam and in the plane of the cross-section. The strengthening system can either be continuous or discontinuous in the longitudinal direction. The use of a continuous pattern may limit the migration of moisture and hence should be considered with caution. In the cross-sectional plane three different configurations may be used these involve the wet lay-up process and are; (i) sheets can be completely wrapped along the cross-section and (ii) wrapped on three sides (U-straps), or bonded on two opposite sides (wet lay-up or re-cured strips). The first pattern is obviously the most efficient, and is typically adopted for shear strengthening columns; however, it is impractical for strengthening beams in the presence of an integral slab. Strengthening on three sides is less efficient because the effective bond length needs to be developed from the free ends, while strengthening only on the two side faces is the least efficient system due to the development of bond length from two free plate ends. Pre-cured plates cannot be bent around corners and can only be bonded to the sides, Zeris [226]. However, pre-fabricated L-angles specifically suited for shear strengthening are also manufactured, Basler et al. [22], Meier et al. [159].

Several different approaches have been used to predict the shear strength of FRP-strengthened RC beams. These include the modified shear friction method, Deniaud and Cheng [56], Deniaud and Cheng [57,58] the compression field theory, various truss models including Mörsch's truss analogy, El-Refaie et al. [63], Ibell and Silva [110], and the design guidelines have adopted the design approach, Maruyama and Ueda [229], *fib*.(2001), ISIS Canada [111], ACI [3], Concrete Society [46,47].

8.6. Bonding FRP composites onto RC structures

The FRP composite plate material used for the upgrading of RC structures is generally the high-modulus CFRP, AFRP (Kevlar 49) or GFRP composites and these composites will be fabricated by one of three methods, namely:

- the pultrusion technique, Hollaway [95], Starr [201]
- the rigid fully cured FRP pre-preg plate, Hollaway [95]
- the cold-melt FRP pre-preg/adhesive film placed onto the structural member, both components are cured and compacted simultaneously on the site structural member under an elevated temperature and pressure; the description of this technique is given in Hollaway [95].
- The wet lay-up process, Hollaway and Head [90].

The first two methods are bonded onto the degraded member with cold cure adhesive polymer. The drawback to these systems is that the plate material cannot be reformed to cope with any irregular geometries of the structural member.

The third method is superior to the pre-cast plate and cold cure adhesive systems as the site compaction and cure procedure of the pre-preg and film adhesive ensures a low void ratio in the composite and an excellent join to the concrete. The drawback to this method, currently, is the cost; it is about twice as expensive as the first two methods. Currently, the preferred manufacturing system by the civil engineering contractors is either the first or second methods.

A two part, cold cure epoxy adhesive is used to bond the plate onto the substrate on site. The bonding operation of the FRP plate to the RC beam has been described by Hutchinson [108].

The cold cured adhesive is the Achilles heel of the plate bonding system particularly if the cure is at a low ambient temperature without post cure, see Section 2.2 (Polymerisation) and Section 2.4.2 (Durability).

Anchoring techniques such as steel bolting and the use of bonded FRP U-shaped channels or jackets at the end of the beam and/or at intermediate locations have been developed to limit de-bonding failures, Quantrill et al. [182,183]. However, the bolting methods must ensure that no damage to the internal steel rebars or the FRP composite occurs during the fabrication operation. The use of FRP anchor spikes has been proposed by Eshwar et al. [66].

8.6.1. Composite patch technology

Adhesively bonded FRP composite patch techniques have been successfully applied to military aircraft repair. The technique is applied to extend the service life of aluminium aerospace components and the method is now of interest to the civil engineering industry to repair cracked metallic materials.

The fatigue of steel sections and the ultimate fracture under cyclic loading is a problem common to many industries but particularly to aging metallic bridges. The cut and re-weld repair techniques are degenerative and replacement of the fatigued or cracked section is often the inevitable solution, causing significant losses in production time; however, recent research has shown that CFRP patches provide an efficient and a relatively easy to apply repair system. Research has been conducted to investigate the bonding of CFRP patches to reinforce cracked steel sections. Righiniotis et al. [187] investigated the potential fatigue life improvement that may be achieved in using CFRP patches on cracked steel members. A two dimensional steel plate with a crack growing into its thickness direction and a CFRP patch attached through a thin fibre/resin film over the crack mouth was modelled and analysed using the finite element method. Comparisons between the stress intensity factors for the un-patched and the patched plates demonstrated the considerable reductions in these parameters resulting from the repair operation. Hongbo et al. [101] have discussed a numerical method of boundary element analysis to reinforce cracked steel plates. The numerical software programme BEASY was adopted to calculate stress intensity factors, crack propagation and fatigue lives of steel plates and the adhesive layer was simulated as interface elements to connect the patch and steel plate. Their work was validated by the good agreement between the numerical and experimental results. Colombi et al. [43] pre-stressed composite patches and bonded them onto cracked steel sections to reinforce cracked details and to prevent fatigue cracking on steel structural elements. Prestressing the CFRP composite patch introduced compressive stresses that produced a crack closure effect. Furthermore, it modified the crack geometry by bridging the crack faces and so reducing the stress intensity range at the crack tip. The debond crack total strain energy release rate was computed by the modified virtual crack closure technique (MVCCT). A parametric analysis was performed to investigate the influence of some design parameters such as the modulus of elasticity of the composite patch, the adhesive thickness and the pretension level on the adhesive-plate interface debond.

8.7. The mechanically-fastened un-bonded FRP (MF-UFRP) method (power actuated (PA)) fastening 'pins' for fastening FRP composites

The mechanically-fastened un-bonded FRP (MF-UFRP) method for the fastening of pre-cast FRP plates uses many closely spaced steel power actuated (PA) fastening 'pins' and a limited number of steel expansion anchors. The method is rapid and uses conventional hand tools, light-weight materials and unskilled labour; minimal surface preparation of the concrete is required and the technique permits immediate use of the strengthened structure. The method has been developed by researchers at the University of Wisconsin, Madison, USA, Bank [15]. An analytical analysis to predict the strength of concrete beams upgraded with the MF-FRP method has been discussed by Arora [10], Bank and Arora [16]. Bank et al. [19,20] have discussed the strengthening of a 1930 RC flat-slab bridge of span 7.3 m by mechanically fastening the rigid FRP plates using the MF-FRP method.

8.8. FRP composite strips for flexural strengthening of slabs

FRP composites have been used for the flexural strengthening of:

- one-way slabs Erki and Heffernan [65], Seim et al. [194],
- two-way slabs using FRP sheets bonded along the middle of the slab, Ebead and Marzouk [60], Harajli and Soudki [79], or distributed along the slab width Mosallam and Mosalam [166], Limam et al. [141], and
- two-way slabs with pre-stressed sheets, Longworth et al. [139].

The interfacial behaviour between the FRP composite and the concrete slab is one of the most important issues as this often controls the failure mode of the strengthened member, Elsayed et al. [64]. If FRP sheets are bonded to the tension face of the slab it is likely that they will cover the whole surface whereas strips would cover only part of its surface. However, with the former method it is difficult to check the quality of the bond and, furthermore, free movement of moisture from the slab is prevented thus increasing the risk of degradation of the bond. Wide strips are possible as they offers a larger contact area between the slab and the reinforcement compared to the narrow strips thus reducing the risk of de-bonding of the strips.

There are a limited number of theoretical studies on FRP-strengthened RC slabs compared to those on beams but the finite element analysis has been one of the most effective numerical methods for modelling their behaviour. However, finite element modelling of laminated beams is difficult because the thickness of the laminates is small compared to the other dimensions of the slab; this leads to a significant increase in the number of elements. Mosallam and Mosalam [166], Ebead and Marzouk [60] used the finite element analysis to investigate the structural behaviour of concrete slabs, both un-strengthened and strengthened. The former investigators reported that the FRP systems upgraded the structural capacity of two-way slabs by up to 200%. Failure was preceded by large deformations providing adequate visual warning. Crushing of the concrete was the common failure mode, with localised de-bonding close to the ultimate load. El Sayed et al. [138] modelled the concrete to FRP interface using appropriate finite elements connecting the FRP composite to the concrete. Limam et al. [141], used a simplified laminated plate model to design strengthened slabs as three-layered plates, where the bottom layer represented the FRP strip, the middle layer represented the steel reinforcement, and the upper layer represented the compressive concrete; a full-bond was assumed between the concrete and the FRP composites. Reitman and Yankelevsky [186] developed a nonlinear finite element analysis of a grid representing the slab which was based on yield-line theory for reinforced concrete slabs under various loading conditions. Seim et al. [194] used a beam analogy technique to provide an approximate solution to the overall response of FRP-strengthened one-way slabs. Michel et al. [162] introduced an analytical model to explain the behaviour of RC slabs strengthened by FRP composite materials bonded to the tensile face of the beam; experimental models were compared with those of the analytical model; they found good agreement between the two techniques.

8.9. Near Surface Mounted (NSM) FRP bars

Another technology for increasing the flexural and shear strength of deficient reinforced concrete (and masonry and timber) members is to utilise the Near Surface Mounted FRP bar. This technique requires a groove to be cut into the concrete surface in the direction for flexural or shear upgrading. The groove is then filled half-way with adhesive epoxy paste and an FRP rod of circular or rectangular cross-section is placed into it and lightly pressed into the paste. This forces the paste to flow around the rod and the sides of the groove. The groove is then filled with more paste and the surface is levelled to that of the concrete, De Lorenzis et al. [55].

The advantages of using NSM FRP rods compared with using EPB laminates are:

- There is no preparation of the concrete surface apart from removing the sawn debris.
- The possibility of anchoring the rods into adjacent members.
- There is minimal installation time.
- The rods are protected from the external environments in that they are completely surrounded in adhesive paste. This implies that concrete structures which have alkaline and other salts in the cements do not attack the paste, therefore, the rods will not be affected by the alkaline initiated corrosion in a concrete environment.

Bond is the first issue that needs to be addressed, since it is the means for the transfer of stress between the concrete substrate and the FRP reinforcement. If manufactured by the pultrusion method the surface of the rods will be smooth and they must be roughened to improve the bond strength. There are two main techniques for surface roughening, namely, CFRP sandblasted, GFRP deformed methods or in conjunction with the pultruded technique to place a FRP composite spirally wound onto the external surface of the bar.

De Lorenzis and Nanni [52] have shown that when NSM round bars are used to enhance the shear capacity of RC beams an increase in capacity as high as 106% in the absence of steel stirrups, and a significant increase also in the presence of internal shear reinforcement. One of the observed failure modes was de-bonding of one or more FRP bars, associated with concrete diagonal tension. This mechanism can be prevented by providing a longer bond length by either anchoring the NSM bars in the beam flange (for T-beams) or using 45° bars at a sufficiently close spacing. When de-bonding of the bars is prevented, splitting of the concrete cover of the longitudinal reinforcement may become the controlling mechanism due to the fact that, unlike internal steel stirrups, NSM rods are not able to exert any restraining action on the longitudinal reinforcement subjected to dowel forces.

De Lorenzis et al. [53], El-Hacha and Rizkalla [62] have shown through experimental test results that NSM reinforcement can significantly increase the flexural capacity of RC elements but bond may be the limiting factor on the efficiency of this technique as it is with externally bonded laminates. El-Hacha and Rizkalla [62] compared the strengthening of reinforced concrete T-beams with identical NSM strips and externally bonded plates, They obtained a strength increase of 4.8 times higher in the first case, due to an early de-bonding failure of the external FRP compared to a tensile rupture of the NSM strips.

A review of the NSM reinforcement technique has been given by De Lorenzis and Teng [54].

8.10. The rehabilitation of steel beams by the technique of external plate bonding

Corrosion, fatigue and a lack of proper maintenance are possibly the major causes for steel bridge deterioration. In addition, many of the bridges require upgrading to carry present day traffic requirements. Clearly, a repair or a rehabilitation option should be considered before a decision is taken to replace a structure, since the cost to undertake the former is far less than the cost of replacement. In addition, repair and rehabilitation will invariably take less time and will reduce the disruption to traffic and commerce.

The superior mechanical and physical properties of CFRP composites make them excellent candidates for the repair and rehabilitation of steel structures compared to many other civil engineering materials. CFRP have excellent mechanical properties, namely, typical tensile and stiffness values of 1.5 GPa and 220 GPa, respectively, for the high stiffness material and 1.5 GPa and up to 420 GPa, respectively, for the ultra high stiffness material; the latter composite is invariably used for the upgrading of steel members. These materials have not been utilised to upgrade metallic structures to the same extent as they have been for reinforced concrete structures and until relatively recently only a limited amount of research had been conducted on the application of these materials to metallic structures, but this situation is now changing, Mertz and Gillespie [152], Mays [155], Mosallam and Chakrabarti [154], Tavakkolizadeh and Saadatmanesh [206], Luke and Canning [147], Luke and Canning [148], Photiou et al. [179], Schnerch and Rizkalla [192].

In principle, the high tensile strength and elastic modulus of carbon fibre polymer (CFRP) composites make them ideal candidates for upgrading steel structures, though it is necessary to appreciate the possible limitations associated with their mechanical properties, their interaction with the steel substrate and their long-term behaviour in harsh environments. The high-modulus (H-M) CFRP composites have stiffnesses of the same order as that of the steels and therefore substantial load transfer can only take place after the steel has yielded. The ultra high-modulus (UH-M) CFRP composites (see Hollaway and Head [90]), can have stiffness values in excess of 600 GPa, and for the upgrading of steel beams the UH-M pitch precursor (manufactured by the cold-melt factory-made pre-impregnated fibre with a compatible film adhesive as opposed to the polyacrylonitrile precursor method for the H-M CFP composite which is manufactured by the pultrusion method and uses the cold cure adhesive for bonding) for the manufacture of CFRP pre-pregs at about 60% FVF are used; these composites when utilised in construction would be manufactured to have moduli values of about 400 GPa. Consequently, the stiffness of this material will be of the order of twice that of the steel; the load transfer to the composite will then commence to take place before the steel has yielded. With the high stiffness moduli values the strain to failure of the UH-M carbon fibres are very low, less than 0.4% strain depending upon the modulus of elasticity value. Photiou et al. [179] have shown that the adhesive film with the pre-preg composite fails at higher ultimate loads compared to the cold cure adhesive/pultrusion CFRP plate. Finally, the successful strengthening of steel structures with FRP materials is completely dependent upon the quality and integrity of the composite/steel joint, and the effectiveness of the adhesive used. The integrity of the joint is highly dependent upon the preparation procedures for bonding the FRP plates onto, possibly corroded, steel members. Consequently, the surface for bonding must be carefully cleaned and be free from rust and scale and any pitting must be levelled using an epoxy paste. The beam surface should then be shot blasted prior to the bonding operation. If an aluminium structure were to be upgraded its surface pre-treatment would include etching and anodising procedures; these operations cause chemical modification to the surfaces involved. The performance of the adhesive joint is directly related to the successful application of the pre-treatment and this in turn depends upon the quality of the surface characteristics of the substrate in terms of topography and chemistry.

A significant advantage of using a film adhesive with a compatible pre-preg is that the curing operation of the composite material and the adhesive resin is undertaken on site in one operation under an elevated temperature of 60° C for 16 h (or 80° C for 4 h) and a pressure of 1 bar. This system was used to fit CFRP composites to an historic building, Garden and Shahidi [73], but to the knowledge of the author it has not yet been employed to upgrade a bridge structure constructed of traditional structural materials.

8.11. The retrofitting of RC columns by using uni-directional FRP composites

During the last decade many experimental investigations have been conducted on retrofitting FRP composites to concrete columns. Experimental tests on concentrically loaded concrete specimens retrofitted with FRP composites to observe the strength and deformation capacity enhancement in pure compression have been undertaken by Samaan et al. [190], Xiao and Wu [222], Harries and Kharel [80], Lam and Teng [133]. These tests revealed that the behaviour of FRP-confined concrete substantially differs from that of steel confined concrete due to differences in constitutive behaviour of the two materials. The level of confinement for the FRP-wrapped concrete is proportional to the axial load up to the point where FRP ruptures and failure occurs in a sudden and brittle manner, whereas for the steel confined concrete the confining stresses are proportional to the applied axial load up to the yielding of the steel. Beyond yielding of the transverse steel reinforcement, confining stresses remain approximately constant. Other studies have concentrated on FRP retrofit of deficient RC columns for seismic strengthening, Saadatmanesh et al. [189], Seible et al. [193], Sheikh and Yao [197], Iabucci et al. [109], and Xiao and Ma [221]. It has been shown that FRP retrofit can significantly improve the lateral deformation capacity of columns keeping the plastic hinge regions intact even at large deformation cycles, Tastani and Pantazopoulou [211].

FRP-confined concrete models have been developed; extensive reviews of the literature on FRP-confined concrete have been investigated by Lam and Teng [132], Teng et al. [207]. Most of these models are empirical in nature and employ best-fit expressions. Other analytical models Binici [27], Kazunori et al. [120], Spoelstra and Monti [200] define the axial and lateral stress–strain relationships of concrete for different levels of confinement. By matching the expansion of concrete to the straining of the jacket, FRP-confined concrete response is obtained from a family of active confined concrete curves. The advantages of these latter models include:

1. The flexibility of introducing different material models for the jacket,
2. The ability to directly define failure of the jacket from the transverse jacket strains.

Although these models have been shown to estimate the axial response of FRP-confined concrete accurately, they have not been used widely for the analysis of columns subjected to combined axial loads and bending moments.

Following the disastrous Hyogoken–Nanbu earthquake which occurred in 1995, Japan has concentrated considerable effort into the studies of economically viable seismic retrofit systems. Of particular concern is the seismic performance of structures originally designed for gravity loads only. RC systems that were not designed for seismic loads can have inadequate ductility and a lack of robustness. Seismic upgrade of these structures can have profound economic and social implications.

9. The FRP rebars used to reinforce concrete beams and slabs

Steel rebars are protected by the high alkalinity (pH 12.5–13.5) of the concrete and are physically protected by the surrounding concrete cover against corrosion. When the structural members are exposed to aggressive environmental combinations of moisture, temperature and chlorides the alkalinity of the concrete is reduced; this combination of attack together with freeze–thaw and de-icing salts on the steel will result eventually in the corrosion of the steel reinforcement and a loss of structural serviceability. To overcome these corrosion problems the use of FRP composite rebars would be an advantageous option, see Section 5.1.3, for the in-service properties of composites.

Durability of FRP rebars is not a straightforward topic; it tends to be more complex than the corrosion of steel reinforcement, because the degradation of the material is dependent upon the components of the composite material. Furthermore, the types of rebars available on the market are various and the commercial products are improving with time. Different fibres are characterised by different behaviour under high temperature, environmental effects and long-term phenomena. In addition, concrete could be an unfavourable environment due to alkali and moisture absorption. Many studies have been carried out on the durability of FRP bars, Balazs and Borosnyoi [14], Katz [119], Karbhari [117], Micelli and Nanni [153], Nkurunziza et al. [175], Uomoto [217,218]; however, there are still many aspects to be investigated in order to provide reliable design rules to be implemented in codes of practice.

The geometric cross-sections of FRP rebars are typically square and rectangular and they are generally manufactured by the pultrusion process from continuous carbon, glass or aramid fibres embedded in vinyl-ester or epoxy matrix. Products of the pultrusion technique have smooth surfaces and these must be modified to improve the bond characteristics between the concrete and the rebar. The improvements in bond characteristics are effected by forming:

- Ribbed bars – manufactured from a combination of a pultrusion and compression moulding method.
- Sand-blasted bars – manufactured by applying sand blasted finish to the pultrusion.
- Spirally wound and sand coated bars – manufactured by spirally winding the pultrusion rod with a fibre tow sand coated.

Other systems for improving bond between the FRP composite and concrete are given in Pilakoutas [180].

The advantages and disadvantages of the use of FRP rebars are given in Hollaway [95], the major difficulty with thermosetting polymer composites on site is that they cannot be reshaped to form hooks or angles for end anchorage; specially shapes rebars are required to be manufactured in the factory. For site shaped rebars made from thermoplastic polymer material is currently used and by the application of heat the bar may be shaped into 90° or 180° bends; sufficient bond length must be provided between the bars from the thermosetting and thermoplastic materials. Composite materials have now become a strong alternative/competitor to steel rebars.

Claims have been made by some researchers, Bank and Gentry [17], Bank et al. [18], Sen et al. [195], that GFRP rebars could be susceptible to the high alkaline environment of concrete where the pore solution concentration has a pH value of 12.5–13.5. Long-term full size tests undertaken by ISIS Canada on RC structures in excess of 10 years have shown that the GFRP flexural tension reinforcement is durable and compatible with concrete, Mufti et al. [169–171].

A state-of-the-art paper on the durability of FRP rebars has been written by Ceroni et al. [36].

The American Concrete Institute (ACI) in 2008 [5,6] published two new guides for concrete reinforcement: (i) specification for construction with fibre-reinforced polymer (FRP) reinforcing bars AND (ii) ACI 440.6-08: specification for carbon and glass fibre-reinforced polymer (FRP) bar materials for concrete reinforcement.

10. The hybrid structural member in new construction

The hybrid structural system consisting of FRP composites and traditional materials such as concrete and/or steel, optimally combined, are currently a major focus for the use of the composite materials in new construction; the successful applications of these systems requires that three criteria should be met, these are:

(i) Cost effectiveness in terms of the most advantageous combination of whole-life cost and of high quality and performance.
(ii) The material should ideally be used in areas subjected to tension, (for instance, in wrapping columns and on the tension soffit of beams).
(iii) The fire resistance should not be critical, (for instance, where the structure is in an open space (e.g. bridges) or the FRP is not required to make any contribution to structural resistance during a fire.

The first two criteria are met when composites are used in combination with other materials to form hybrid structures. The aim of the designer of these structures should be to optimally combine the FRP with traditional structural materials to create innovative structural forms. With respect to fire, Advanced Composites Group Ltd. (ACG), Derbyshire, England, UK have launched a new a phenolic resin system (MTM 82S-C), available as a pre-preg which has been designed to offer outstanding fire performance to mass transit, industrial, and construction applications. ACG claims that the new pre-preg has excellent mechanical properties in combination with exceptional fire performance, where the operating temperature is within the range −55° C to 80° C. The laminates exhibit extremely low fire propagation and surface spread of flames, together with low smoke and toxic gas emission. Consequently, the product meets the requirements of: the British Standards BS 476 Parts 6 and 7, and BS 6853 Cat 1a, the French Standards NF P 92-501 Rating M1 and NF F

16-101 Rating F1, and the German DIN5510 Ratings S2, SR2 and ST2.

During the last decade the Universities of California, San Diego, USA, Southern Queensland, Australia, Surrey, UK, Warwick, UK, the Hong Kong Polytechnic University, China and EMPA, Switzerland have focussed upon hybrid systems that combine advanced composites with conventional materials, currently, mainly for bridge constructions.

Examples include: concrete-filled FRP tubes as columns and piles, FRP cables and FRP composite/concrete duplex beams. Hybrid systems previously mentioned are the upgrading of structures with composites, the rebars to form reinforced concrete and the bridge decks supported by the superstructure of the bridge.

10.1. Examples of hybrid systems

(1) In a reinforced concrete structural beam member the region below its neutral axis is wasteful of material due to the weakness of the concrete in tension; it merely holds the reinforcement in position and protects it from aggressive environments. Furthermore, this area of concrete adds weight to the beam unnecessarily and hence increases the foundation size. This region of the beam could be substituted by a FRP composite structural unit; composites with high fibre volume fractions have high specific strength and stiffness. Moreover, it has been shown in Section 5.1.3 that GFRP, CFRP and AFRP composites possess excellent in-service properties particularly durability; therefore, they require only minimum maintenance; depending upon the requirements of the structural system the composite may require fire protection.

An innovative hybrid rectangular beam cross-section composed of a low-cost construction material, (namely, concrete) placed in its compressive region and a high specific strength/stiffness FRP composite situated in the tensile region was presented by Triantafillou and Meier [213], Deskovic and Triantafillou [59], Triantafillou [214], Canning et al. [31], Hulatt et al. [104]. This system resulted in a new concept for a light-weight structural member which was corrosion free with excellent damping and fatigue properties. This system was extended to form a composite/concrete duplex beam for both a standard rectangular and a Tee beam cross-section, Hulatt et al. [102], Hulatt et al. [103]; the webs of both sections were constructed as a GFRP plate or as a sandwich plate section and a CFRP plate was incorporated into the soffit of the beam. Further developments of this beam system have been discussed in Hulatt et al. [105,106]. The VTM260 series epoxy resin, glass and carbon fibre pre-pregs supplied by Advanced Composites Group (ACG), Heanor, UK were used in this research at the University of Surrey. Using this hybrid structural beam system and ACG's VTM264 variant epoxy/carbon fibre pre-preg material NECSO Entrecanales Cubiertas, Madrid, Spain undertook a R&D project and developed a 'Duplex' beam element; this system was utilised as an advanced composite motorway bridge construction on the highway at Cantabrico in Spain; Hollaway [95] shows the completed bridge. The production of these first advanced composite bridge beam elements has demonstrated that utilising proven composite design principles, civil engineering structural elements can be produced that offer significant benefits over traditional reinforced concrete designs. There are, in the offing one or two further designs for bridges in Spain using this method of construction. In recognition for this development and the pre-preg composite technology, ACG was awarded the JEC Composites Award 2005 for construction, reinforced plastics (2005). Khennane [124] has taken this idea a stage further with the rectangular section member manufactured from pultuded GFRP composite and a CFRP laminate in the tension zone. The concrete in the compression area of the section is manufactured from high-performance concrete (HPC).

(2) Concrete-filled steel tubes have been used as structural columns but these have been largely superseded by hybrid FRP columns consisting of an FRP tube filled with concrete with or without internal reinforcement, Mirmiran et al. [128], Fam and Rizkalla [68], Mirmiran [165], Xiao [223], Mirmiran et al. [164]. Double-skin hybrid FRP columns consisting of two concentric FRP tubes and the space between them filled with concrete have been also been studied Fam and Rizkalla [69]. The advantages of the concrete-filled FRP tubes over the concrete-filled steel tubes include light-weight and corrosion resistance. However, building columns manufactured from concrete-filled FRP tubes do have a number of disadvantages, these include brittle failure in bending, difficulty with connections to beams, an inability to support substantial construction loads and poor fire resistance; this last resistance is not significant with respect to bridge columns. Teng et al. [210] has suggested a new form of hybrid column to overcome the disadvantages discussed above. Teng's column consists of an outside FRP tube and a concentric steel tube inside; the annulus is filled with concrete. The fibres are oriented mainly in the hoop direction in the FRP tube thus providing confinement to the concrete for enhanced ductility and additional shear resistance. The stated aim of the new column is to achieve a high-performance structural member by combining the advantages of the three constituent materials and the structural form of the columns which are mentioned above. The column is relatively easy to construct and is highly resistant to corrosion and earthquakes. Clearly, the section form can also be employed as a beam by moving the inner steel tube towards the tension side, Teng et al. [210] has illustrated these sections.

11. The steel-free deck system

Extensive research was conducted in Canada at the end of the 20th century and into the beginning of the 21st century has led to concrete deck slabs of bridges that can be entirely free of any tensile reinforcement, Mufti et al. [230]. By omitting the tensile steel rebars, thus avoiding corrosion and major bridge deck deterioration, and replacing them with external steel straps positioned at the top flange of the superstructure girder for lateral restraint would lead to an increase in the service life of bridges. Compressive member forces in the deck slab are developed under increasing traffic loads which will eventually reach a magnitude where the tensile stresses will cause the concrete to crack. It has been shown that the wide cracks do not pose any danger to the safety of the structure, however, they are generally unacceptable to bridge engineers; aesthetically cracks are unpleasant. The cracks are controlled by a mesh of nominal GFRP bars. Once this happens the deck will resist traffic loads through arching action thus causing compressive membrane action and cracking on the soffit of the deck slab. The first application of the second-generation slab was on one span of the 10-span Red River Bridge on the North Perimeter Highway in Winnipeg, Canada, Klowak et al. [113]. Gordon and May [74] have stated that the utilisation of 'steel-free' decks may be of use in Europe initially as temporary and accommodation bridges. Mufti et al. [169] have illustrated the arching action. Klowak et al. [129] have discussed a second generation steel-free slabs for bridges rehabilitation. Mufti and Neale [173] have written a state-of-the-art paper on FRP and SHM applications in bridge structures in Canada; a discussion on steel-free deck systems is included in their paper.

12. The future direction for FRP composites in the construction industry

12.1. Introduction

The simplest way to introduce a new material into construction and thence become familiar with its potentialities is to replace the traditional material with the new one. This approach was employed by the building industry in the late 1960's, clearly this approach is limited as the building systems which had been developed for the traditional materials had been evolved and designed over many decades. The innovation in FRP composite materials for the civil/structural engineering commencing in the 1980's made rapid advances as this paper has illustrated. The main interest has been in the combination of FRP composites with the more primary traditional materials and examples of this have been given but as yet fewer studies have concentrated on 'all FRP' structural elements for bridges and buildings. However, this area is expanding with the introduction of FRP deck systems for bridge construction and the construction of FRP footbridges and small span highway bridges. The main advantages of FRP composite materials have been discussed in Part A with a special mention to its very satisfactory durability property compared with traditional materials. The main disadvantages have been alluded to as the fragile nature of the composite material compared to the traditional construction materials and the relative inexperience of some of the FRP designers and general construction contractors which has resulted in high safety factors being applied at the time of the design. The high cost of the material is another factor which prevents its greater use but when the whole-life cost of the construction and long-term durability of the material is taken into account, the cost of using this material is generally cheaper than that of the traditional materials. It has already been mentioned that the first achievements in construction utilising this material dates back to the beginning of the 1980s, indeed the building industry dates back to the beginning of the 1970s when the structures were manufactured from hand lay-up polyester and a randomly orientated glass fibre; these structures, as reported in Section 6, are showing minimal degradation. Hollaway [91] and Market Development Alliance of the FRP Composites Industry, Global FRP Use for Bridge Applications, (2003), have given state-of-the-art reviews of the use of FRP composites up to that time.

12.2. Future directions

The future directions of the utilisation of FRP composites in the construction industry and the methods of monitoring structures throughout their lives possibly in hostile environments depend upon innovative ideas. Some suggestions are discussed in the following items.

(i) There have been some interesting ideas put forward over the last three decades which have shown how FRP composites, involving innovative ideas, can be used where traditional materials are not capable of the same function. In 1987 Professor Urs Meier of EMPA, Switzerland suggested that composite materials offered the opportunity to build a bridge across the Straits of Gibraltar, this innovative idea captured the imagination of both the international bridge and composites engineering communities. Steel cables would be limited in this application as they could not have supported their own weight when the bridge span approached 7 km in length; to span the Strait of Gibraltar would require a minimum central span of 8.4 km. Meier [160], showed that the use of carbon-fibre-reinforced polymer components for the bridge deck and cables would allow a significant increase in the limiting span of such a bridge and would have permitted the construction of the bridge over the Strait of Gibraltar.

Another idea put forward by Schlaich and Bleicher [191] to show the potential of CFRP composites was a stressed-ribbon bridge with carbon fibre ribbons. The ribbons were anchored at the abutments on both sides of the bridge. The ribbons are generally made from steel plates or steel cables which are covered and stabilised by open-jointed concrete slabs but Schlaich and Bleicher [191] replaced the steel units by CFRP ribbons. By using CFRP ribbons of low specific weight and high specific tensile strength will allow longer span bridges with smaller cross-sections to be built compared with those of steel.

(ii) The first carbon-fibre tendon cable-stayed vehicle bridge in the world was the Storchenbrücke (Stork Bridge) at Winterthur in Switzerland with twin spans of 63 m and 61 m. Two of the 24 stays are carbon composite tendons put in place to establish their viability and long-term behaviour in this application; the other 22 stays are steel. The bridge-deck spans both a river and a railway station crossing the major east-to-west axis of the Swiss Federal Railway Network, was the world's first bridge to use carbon stay cable technology, Meier [161]. Due to its low self-weight, carbon stay cables are a promising solution for ultra long span bridges. Their extremely high fatigue resistance and the fact that carbon is non-corrosive are further advantages of this type of cable; however, special care should be taken when choosing an anchorage system for carbon stay cables.

(iii) The performance of buildings to blast loads is an ever increasing issue; this is clearly evident from the recent world events. Many older buildings such as un-reinforced masonry infill walls have low flexural capacity and a brittle mode of failure and therefore they will have a low resistance to out-of-plane loads. To improve the latter's resistance in the past FRP composites have been retrofitted but one drawback to using these materials is their lack of ductility at the ultimate state. Casadei and Agneloni [35] have investigated a hybrid system that couples together the high-strength of the FRP systems with the ductility of polyurea resin that can elongate up to 400%. The authors claim that the system allows different layers of FRP and Polyurea to

- provide the necessary strength to the infill wall subjected to out-of-plane forces,
- provide a ballistic layer to catch flying debris, and
- guarantee sufficient deformations of the strengthened walls dissipating energy without collapsing completely.

(iv) Barriers to protect airport infrastructure against malicious actions or natural events. Asprone et al. [11] have investigated the design, optimisation and testing of a fence system made from GFRP composites which could protect critical airport infrastructures without disturbing radio-communications. Electromagnetic tests confirmed radio-transparency of the barrier, mechanical tests guaranteed its high structural performances and in situ blast tests confirmed its capacity to withstand blast loads and reduce shock wave effect on protected targets.

(v) The main structural element for structures particularly bridges currently is the pultruded unit which is a cost-effective production method. The use of pultruded structural members has been demonstrated in this paper, for new bridges and buildings and the rehabilitation of concrete and steel structural members. The high quality of the pultruded elements can be assured as they are manufactured in a factory environment under strict quality control to ensure the fibres are positioned correctly as they pass through the heated die and the temperature and pressure of the injected polymer are as specified. Furthermore, transportation costs for FRP pultruded structural elements are lower in comparison to the traditional materials due to their specific weight.

(vi) A self-stressed bowstring footbridge which is based on an elastically curved pultruded FRP cylindrical pipe, the geometry of which is that of an elastica, has been introduced by Caron et al. [34]. A cable is anchored at the ends of the curved member to maintain its shape, and two zigzag stay strings to maintain the curved arches and bows. The energy stored elastically in the bent bows provides the self-stress that generates the required stability for the whole structure; no heavy equipment is required to buckle the curved members.

(vii) Reuse of structural composite materials is high in the waste hierarchy, but may not provide the most practical option for many FRP composite applications. The way in which a material is used, its applications and how it is secured to other components must be considered with a view to deconstruction and reuse at the end of that application's life. The manufacturing process must be examined to identify any possible modifications to enable design for future reuse or recycling. The US Army has commissioned two bridges constructed entirely from recycled consumer and industrial plastics using Axion International Holdings, Inc. N.J., proprietary immiscible blending to create Recycled Structural Composites (RSC). These bridges will be constructed at Fort Eustis, Va, the home of the US Army Transport Corps. The load rating capacities of these bridges is 130 tonnes. This must be seen as an excellent use of waste FRP composite material.

(viii) Cured-in-place-pipes (CIPP) which are designed to reline existing pipe infrastructure without costly excavation, is an innovative use of composites. For example, Reline America (Saltville, Va.) in Amarillo, Texas has recently installed a 600 mm diameter CIPP pipe. CC Technologies Laboratories Inc. (Dublin, Ohio) in conjunction with National Association of Corrosion Engineers, Houston, Texas, (NACE International) has estimated that the direct cost of metallic corrosion in the US is $300 billion (USD) per year. There are significant corrosion costs in metallic pipe technologies, including water and sewer systems, industrial plants and electrical utilities and the corrosion-resistant composites are ideally suited to replace these systems.

(ix) Underwater FRP repair became a possibility following the availability of resins that can cure under wet conditions. However, the application is somewhat controversial because of the uncertainty regarding the continuing corrosion inside the wrap. Nevertheless, several demonstration projects have been completed although the majority of such repairs were undertaken in dry conditions, Sen [196]. The repairs are only durable if the conditions responsible for the original damage are removed. For corrosion damage all chloride-contaminated concrete must be removed. For piles that are half submerged in salt water the cost of such preparation is very high and therefore the pile repair is undertaken in conjunction with a cathodic protection system otherwise the repaired region will corrode again very quickly.

The issues associated with the repair of underwater structures are:

- *Preparation of surface of structure:* The repair of the pile using FRP composite is part contact-critical, to allow expansion due to the corrosion, and part bond-critical to allow for strengthening of the pile.
- *Cleaning corroded steel:* There are guide lines for the repair of deteriorated concrete, provided by the International Concrete Repair Institute, Concrete Repair Manual [45].
- *Access to pile,* The success of the repair requires easy access to the entire region to be wrapped.
- *Environmental conditions:* The temperature wind and tide will interfere with the placement of the wrap and can adversely affect the quality of the repair.
- *Repair region:* The steel rebars which are exposed to the splash zone of the pile are particularly likely to corrode due to a combination of deposit of salt onto the pile surface during the wet/dry cycles of the tide.

(x) Structural health monitoring (SHM), in recent years, has attracted significant interest from academia, government agencies, and industries involved in a diverse field of disciplines including civil, marine, mechanical, military, aerospace, power generation, offshore and oil and gas. The aim of SHM is to detect damage initiation and to subsequently monitor the development of this damage using structurally-integrated sensors in order to provide early warning and other useful information for successful intervention to preserve the structural integrity of the host. Optical fibre sensors are widely used for SHM applications to measure strain, load, displacement, impact, pH-level, moisture, crack width, vibration signatures, and the presence of cracks by modifying a fibre so that the quantity to be measured modulates the intensity, phase, polarisation and wavelength or transit time of light in the fibre. Optical fibre-based sensors such as fibre Bragg gratings (FBG), intensiometric and polarimetric-type sensors and those based on interferometric principles (e.g., Fabry–Perot) have been shown to offer specific advantages in their niche area of applications.

The advantages of optical fibre sensing in engineering structures include their insensitivity to electromagnetic radiation (especially in the vicinity of power generators in construction sites), being spark-free, intrinsically safe, non-conductive and lightweight, and also their suitability for embedding into structures. To date, a number of key optical fibre sensors have been reported and their applications for damage detection in FRP composite structures are given in review articles, Kuang and Cantwell [130], Zhou and Sim [227], Kuang et al. [131].

(xi) Nanoparticles can be classified into three categories depending upon their number of nanoscale dimensions, namely, nanoplates, nano-tubes and nano-spheres. Chemically treated layered silicates (clays) which come under the definition of nano-plates can be combined with normal polymer matrix materials to form a nano-composite in which clay layers are distributed throughout the material. Le Baron et al. [135], Ray and Okamoto [184], Utraki [219], have shown that these high aspect ratio clays can alter the properties of a range of thermoplastic and thermosetting polymers by a number of mechanisms, for instance, by improving their mechanical and thermal properties and reducing permeability. As discussed in Part A there exists a number of different properties that can be tested with respect to any material, these can be divided into four categories, mechanical, physical, thermal and durability. The first three can be tested in a more direct manner than those of durability which is time dependent and involves more complex variables. Composite materials are frequently used in close proximity to, in contact with or enclosed in concrete. The high pH value of the cement pore solution generates a problem related to the durability of glass fibre composites which are susceptible to attack from alkali due to the reactivity of the glass fibre itself. Freshly manufactured concrete will have a pH value of 12.8 increasing to 13.4 after 7 days (Li et al. [140]. Carbon fibre composites on the other hand do not absorb liquids and are subsequently resistant to all forms of ingress from alkalis or solvents, Balazs and Borosnyoi [14]. Polymer–organoclay nano-composites are able to reduce the rate of permeability of salt solutions into the polymer, Zhou and Lee [228], Haque et al. [78] and thus delay the attack on the glass fibre from alkali solutions.

Calcium silicate hydrate, a naturally occurring material in cements, is composed of particles little more than a nanometre in size and tools have been developed to manipulate its behaviour at the nanoscale to form stronger and denser concrete. Scientists at the University of Michigan, USA have developed a mixture of carbon nano-tubes and polymer that is very strong and has electrical properties that allow it to act as a sensor skin. The carbon nanotubes and polymer material may be applied to the surface of a structure or, for instance, may be added to a pre-fabricated panel and it would then act to provide environmental protection and constant monitoring of a structure's condition throughout its surface as opposed to a few locations.

(xii) Sustainable materials/environment development The most commonly quoted definition of sustainability which aims to be more comprehensive than most is that of Brundtland [30]: *Sustainable development is development that meets the needs of the present without compromising the needs of future generations to meet their own needs.*

'Sustainability is a process of change in which the exploitation of resources, the direction of investment and institutional change are all in harmony and enhance both current and future potential to meet human needs and aspirations', Lee and Jain [137]. However, the fabrication of the matrix and fibre of composite materials are of primary concern when considering sustainability as the former component is produced from one of the fossil fuels such as crude oil and natural gas and the latter component (viz. glass, carbon or aramid for civil engineering consumption) require high temperatures during production (1400° C for glass; 1200–2400° C for carbon) and in the case of the latter two fibres require petroleum by-products as precursors. It is worth mentioning that the more conventional civil engineering materials also have high environmental cost associated with them. For instance, during the refining process of the manufacture of steel the removal of the oxygen from the mined iron ore is by heating it with coke and limestone to a temperature of about 1600° C in a blast furnace. In the case of the manufacture of cement a combination of the fundamental raw ingredients of cement, namely, calcium, silicon, aluminium, and iron are mixed in a rotating kiln and the burning fuel inside the kiln reach temperatures of 1430–1650° C. Nevertheless, when considering the energy component of the FRP composite and the material resources in isolation it would appears that the argument for FRP composites in a sustainable environment is uncertain. However, such a conclusion needs to be evaluated in terms of the potential advantages of FRP composites in terms of:

(i) The in-service and mechanical properties and in particular its long-term durability (see Part A), when compared with the more conventional materials.
(ii) Its utilisation in conjunction with the conventional materials in terms of rehabilitation of structures, seismic retrofitting to columns, the manufacture of bridge decks and the hybrid structures to form cost-effective structures in terms of whole-life cost and to provide an economic structural system.

(xiii) From the point of view of geopolitical or environmental issues it is clear that the present methods of producing energy are not sustainable. The currently known sources of coal and oil and the magnitude of their deposits are finite, in addition, the price of these fuels continue to rise, this is causing the renewable forms of energy to become more cost-effective and profitable. Wind power is the world's fastest growing energy source (renewable or otherwise) and as a result, the giant rotor blades on the turbines have become the composites industry fastest growing application. During 2007, slightly more than 20,000 MW of wind power was installed worldwide and according to the European Wind Energy Association. (EWEA) this represents a total installed capacity of 94,112 MW, an increase of 31% compared with 2006. This represented about 17,000 turbines, or nearly 50,000 blades. Assuming a 1.5-MW turbine has a typical blade of 36 m length and weighing about 5216 kg this represents nearly 200,000 metric tonnes of composite materials; manufacturing wind turbine blades represents one of the largest single applications of engineered composites in the world. The vast majority of that total tonnage is glass fibre and thermosetting polymers (either epoxy or vinyl-ester polymer); carbon fibre composites are also used as they have a greater strength and are lighter than GFRP composites. The best prospects for large-scale production and net-energy performance remain wind, earth based and space based solar energies. The first two renewable energies face important limitations due to intermittency, remoteness of good ground sources; the space based solar power is limitless. The basic idea of space based solar power structural systems is to place large solar arrays into the intensely sunlit low earth orbit or ideally in geostationary orbit and collect gigawatt of electrical energy and then beam it to earth using either lasers or microwaves. This system would require many large double layer skeletal space structures manufactured from FRP composites to support the solar collectors and reflectors. Hollaway [99] as discussed this idea with particular reference to the hostile environments at low earth, and at geostationary orbits. Solar power has great potential, it is the largest energy source available to mankind for consumption on earth and the rate at which it is used today does not affect the amount that can be used tomorrow. If, however, the solar radiation received today is not trapped and utilised, it will not be available tomorrow, the sun is not an infinite resource and is slowly (albeit very very slowly) running down. Heinberg and Mander [86] have also made an important observation: 'a full replacement of energy derived from fossils fuels with energy from alternative sources is probably impossible over the short term; it may be unrealistic to expect it even over longer time frames'. Nevertheless, whatever systems are used to harness renewable forms of energy, they are providing and will continue to provide ideal opportunities for the FRP composite industry and this industry should take full advantage of these opportunities.

13. Observations

It will be clear from the paper that in the foreseeable future it is likely that the main utilisation of the APC materials will be used in conjunction with the more conventional materials. The combination of these two dissimilar materials will be utilised in such a way within the structure that the benefits, in terms of the mechanical and in-service properties and the economics of the complete system, will be clearly seen. The advantages of FRP composites can be realised from its physical characteristics and their potential in developing structural systems with service lives exceeding traditional materials. The light weight of the composite can result in lower construction costs and increased speed of construction resulting in reduced environmental impacts. In the case of FRP composite materials' high-strength and stiffness characteristics will require less material to achieve similar performance as traditional materials resulting in minimising resource use and waste production. In general, the advantage of FRP composites is its potential to extend the service life of existing structures and to develop new structures that are far more resistant to the effects of aging, weathering, and degradation in severe environments. It has been shown that the use of FRP composites for construction of new structures and rehabilitation of existing structures has increased significantly over the past decades. Due to its advantageous characteristics, FRP composites have been utilised in new construction of structures through its use as reinforcement in concrete, bridge decks, modular structures, formwork and external reinforcement for strengthening and seismic upgrade. Whilst the mechanical advantages of using FRP composites have been reported widely in the literature and the in-service properties have been discussed in Part A, questions remain as to the feasibility of FRP composites within the framework of sustainable environments. It has been mentioned in the last part of Section 12 of this paper that the evaluated of FRP composites in the construction industry must be looked at in terms of the advantageous specific strength and stiffness of the composite, low environmental impact, low human and environmental health risks, sustainable site design strategies and high-performance. In addition, innovations are required to reduce costs of production and minimising environmental impacts. In terms of implementation, the development of codes and standards that include considerations for safety, performance, and sustainability are needed to transfer technology from laboratory to the market.

Based on the above discussion, it is apparent that the area of hybrid structures should be a major research focus in the use of FRP composites in new construction. Within the area of hybrid structures, the aim should be to optimally combine FRP with traditional structural materials such as steel and concrete to create innovative structural forms that are cost-effective and of high-performance. To this end, simple duplications of existing structural systems are often inadequate. Furthermore, Section 12 has discussed the future direction for innovative uses of composite materials for construction and probably the greatest use yet of structural FRP composite is/will be in the area of structural members for the harnessing of energy from Wind, wave, hydroelectric, underground coal gasification, geothermal and solar power.

Appendix A

A.1. Typical mechanical values

See Table A1–A5.

Table A1
Typical tensile mechanical properties of glass, carbon and aramid fibres.

Material	Fibre	Elastic modulus (GPa)	Tensile strength (MPa)	Ultimate strain (%)
Glass fibre	E	69	2400	3.5
	A	69	3700	5.4
	S-2	86	3450	4.0
Carbon fibre				
Pan based fibre				
Hysol Grafil Apollo	HM[a]	300	5200	1.73
	UHM[b]	450	3500	0.78
	HS[c]	260	5020	1.93
Pan based fibre				
BASF Celion	G-40–700	300	4960	1.66
	Gy 80	572	1860	0.33
Pan based fibre Torayca	T300	234	3530	1.51
Pitch based fibres				
Hysol union carbide	T-300	227.5	2758.0	1.76
	T-500	241.3	3447.5	1.79
	T-600	241.3	4137.0	1.80
	T-700	248.2	4550.7	1.81
Aramid fibre	49	125	2760	2.2
	29	83	2750	3.3

[a] High-modulus (American definition is known as intermediate modulus).
[b] Ultra-high-modulus (American definition is known as intermediate modulus).
[c] High-strength.

Appendix B

B.1. Design codes, codes and specifications for the design of FRP composites in structural engineering

In recent years a significant number of design codes and specifications have been published by technical organisations which provide guidance for design with FRP materials for civil engineering. The key publications are listed below.

B.1.1. British and European

- ***Structural Design of Polymer Composites*** Eurocomp Design Code and handbook. Edited by John L Clarke., (1996).
- ***fib Task Group 9.3, FRP Reinforcement for Concrete Structures***, Federation Internationale du Beton (1999).
- ***fib Bulletin 14, Design and use of externally-bonded FRP Reinforcement for RC Structures***, Federation Internationale du Beton [71,72].
- ***'Strengthening Concrete Structures using Fibre Composite Materials: Acceptance, Inspection and Monitoring'*** TR57, Concrete Society, Camberley, UK [46].
- ***'Design Guidance for Strengthening Concrete Structures Using Fibre Composite Materials'***, TR55, 2nd ed., Concrete Society, Camberley, UK [46].
- ***'Strengthening Metallic Structures Using Externally Bonded Fibre-Reinforced'*** Cadei, J.M., Stratford, T.K., Hollaway, L.C., and Duckett, W.G. CIRIA Report, C595, (2004).
- ***Eurocrete Modifications to NS3473 – When Using FRP Reinforcement***, Report No. STF 22 A 98741, Norway (1998).

Table A2
Typical tensile mechanical properties of the three thermosetting polymers used in civil engineering.

Material	Specific strength	Ultimate tensile strength (MPa)	Modulus of elasticity in tension (GPa)	Coefficient of linear expansion (10^{-6}/°C)
Thermosetting				
Polyester	1.28	45–90	2.5–4.0	100–110
Vinyl-ester	1.07	90	4.0	80
Epoxy	1.03	90–110	3.5	45–65

Table A3
Typical tensile mechanical properties of long directionally aligned fibre reinforced composites (fibre weight fraction 65%) manufactured by automated process (the matrix material is epoxy).

Material	Specific weight	Tensile strength (MPa)	Tensile modulus (GPa)	Flexural strength (MPa)	Flexural modulus (GPa)
E-glass	1.9	760–1030	41.0	1448	41.0
S-2 glass	1.8	1690.0	52.0	–	–
Aramid 58	1.45	1150–1380	70–107	–	–
Carbon (PAN)	1.6	2689–1930	130–172	1593	110.0
Carbon (Pitch)	1.8	1380–1480	331–440	–	–

Table A4
Typical tensile mechanical properties of glass fibre composites manufactured by different fabrication methods.

Method of manufacture	Tensile strength (MPa)	Tensile modulus (GPa)	Flexural strength (MPa)	Flexural modulus (GPa)
Hand lay-up	62–344	4–31	110–550	6–28
Spray-up	35–124	6–12	83–190	5–9
RTM	138–193	3–10	207–310	8–15
Filament winding	550–1380	30–50	690–1725	34–48
Pultrusion	275–1240	21–41	517–14,448	21–41

Table A5
Typical tensile mechanical properties of glass fibre/vinyl-ester polymer (compression moulding – randomly orientated fibres).

Fibre/matrix ratio (%)	Specific weight	Flexural strength (MPa)	Flexural modulus (GPa)	Tensile strength (MPa)	Tensile modulus (GPa)
67	1.84–1.90	483	17.9	269	19.3
65	1.75	406	15.1	214	15.8
50	1.8	332	15.3	166	15.8

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